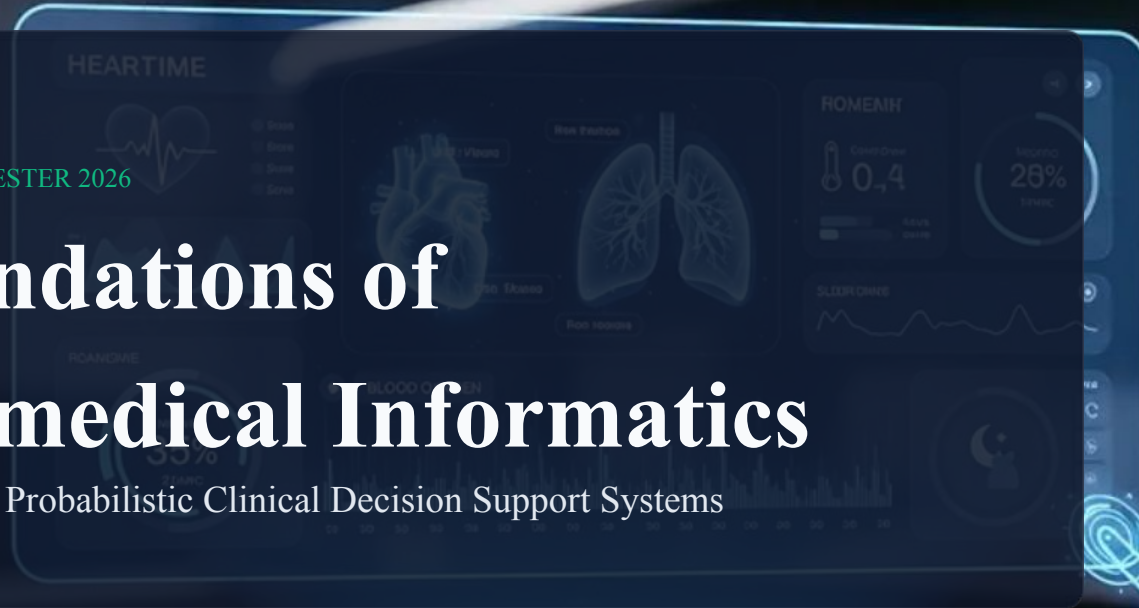


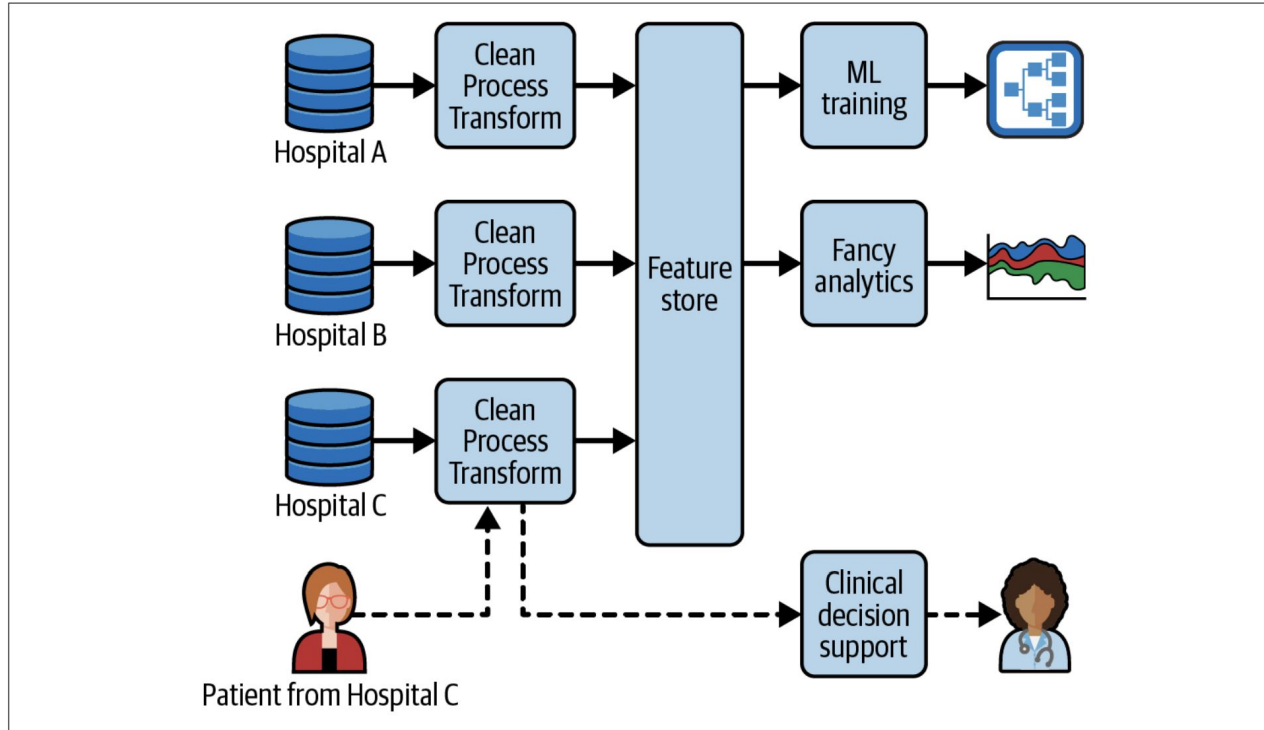
WINTER SEMESTER 2026

Foundations of Biomedical Informatics

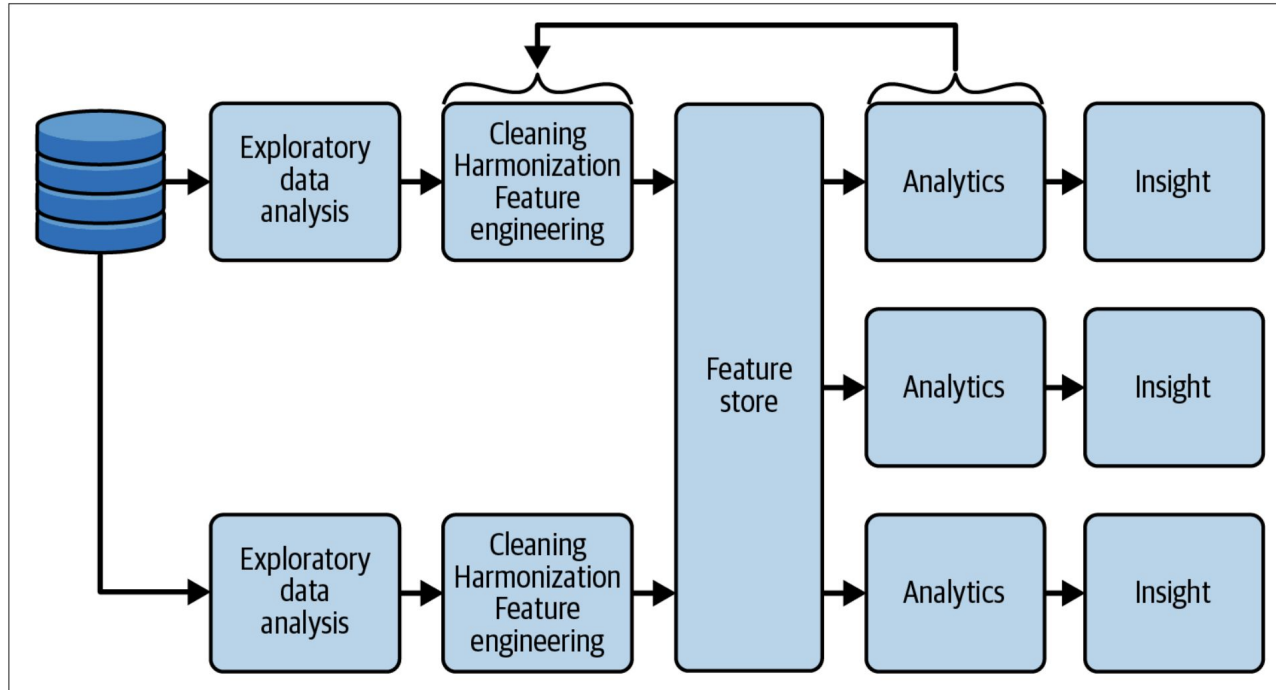
Lecture 11: Probabilistic Clinical Decision Support Systems



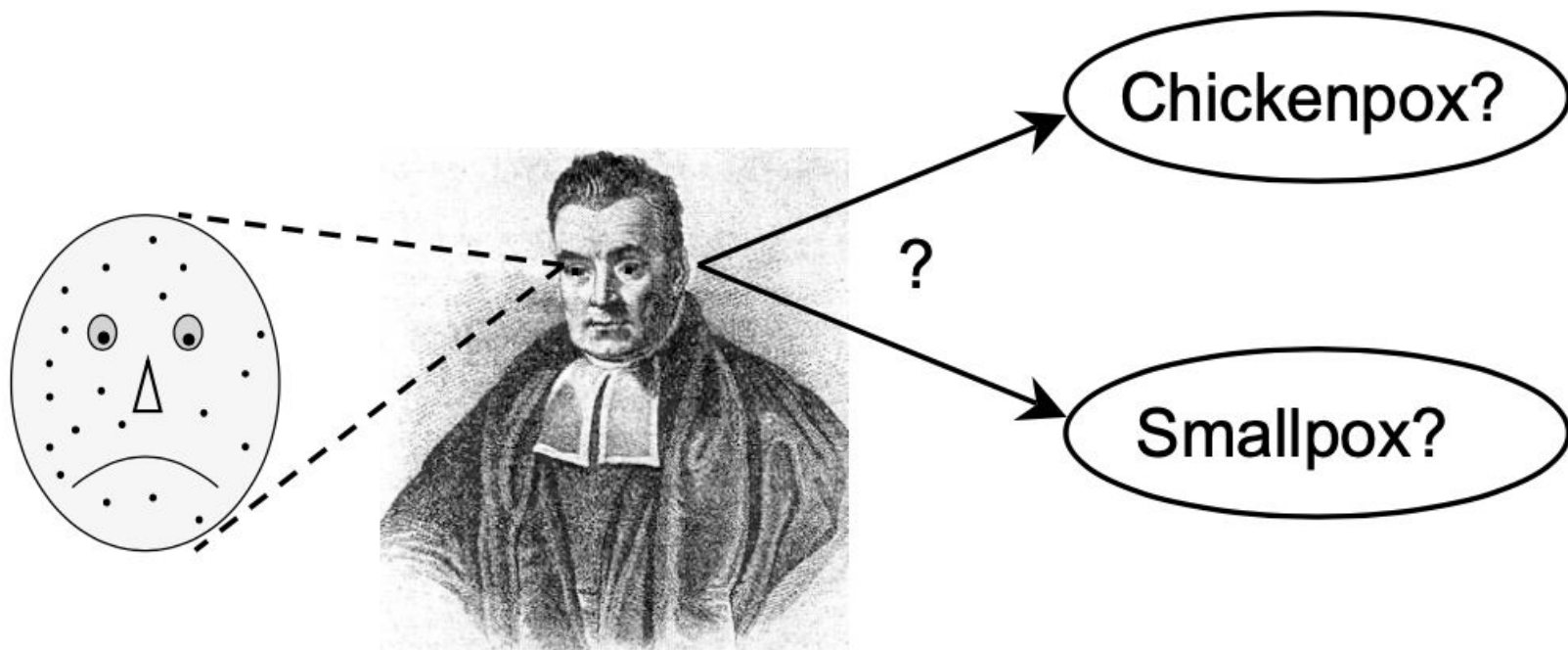
Interoperable Decision Support



Interoperable Decision Support



Bayes Rule



We know,

$$p(\text{spots}|\text{smallpox}) = 0.9.$$

$$p(\text{spots}|\text{chickenpox}) = 0.8.$$

**When You Hear Hooves, Think Horse,
Not Zebra**



Likelihood

$$p(\text{spots}|\text{smallpox}) = 0.9.$$

Likelihood of smallpox = probability of spots given smallpox

$$p(\text{spots}|\text{chickenpox}) = 0.8.$$

Likelihood of chickenpox = probability of spots given smallpox

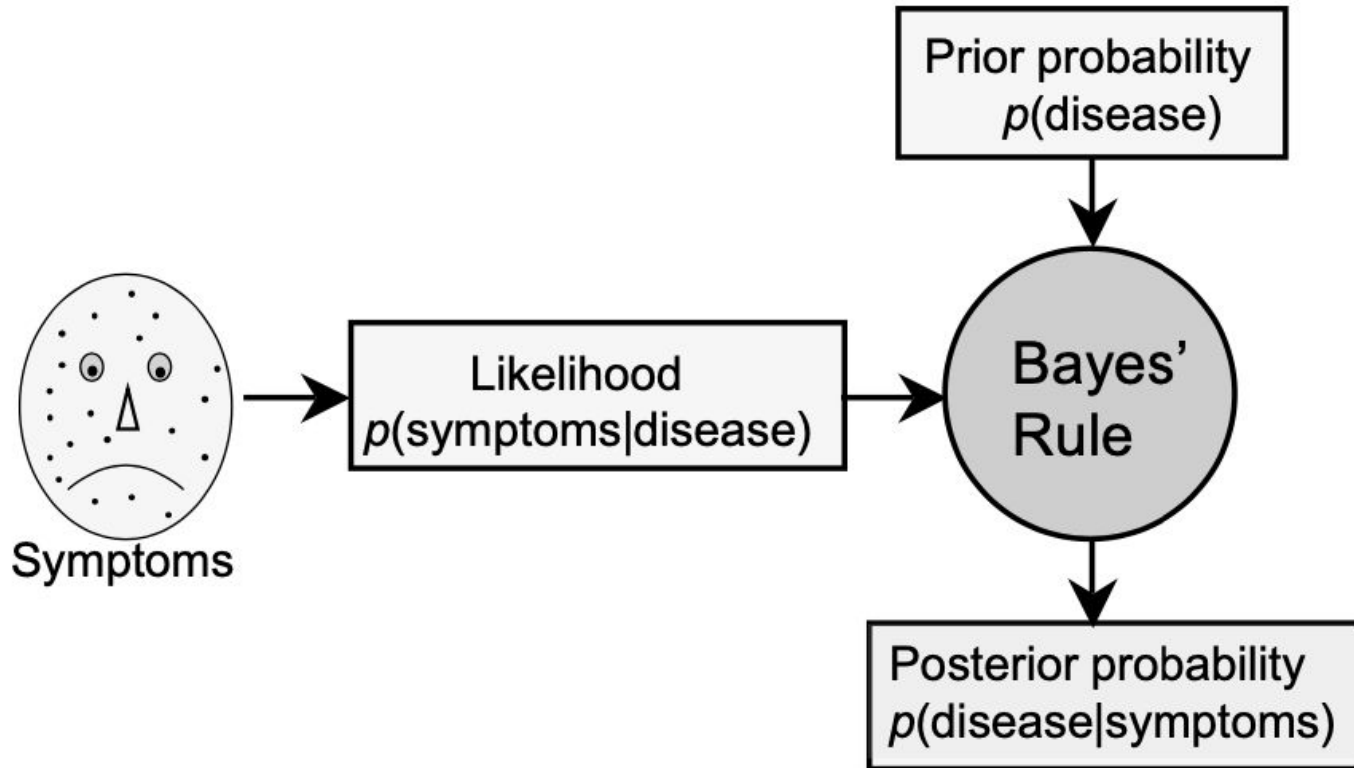
Beware the confusion in language!

Maximum Likelihood Estimate

$$p(\text{spots}|\text{smallpox}) = 0.9. \quad > \quad p(\text{spots}|\text{chickenpox}) = 0.8.$$

Statistical models that work by maximizing the value of likelihood is known as maximum likelihood estimate (MLE)

Bayes Rule in Machine Learning

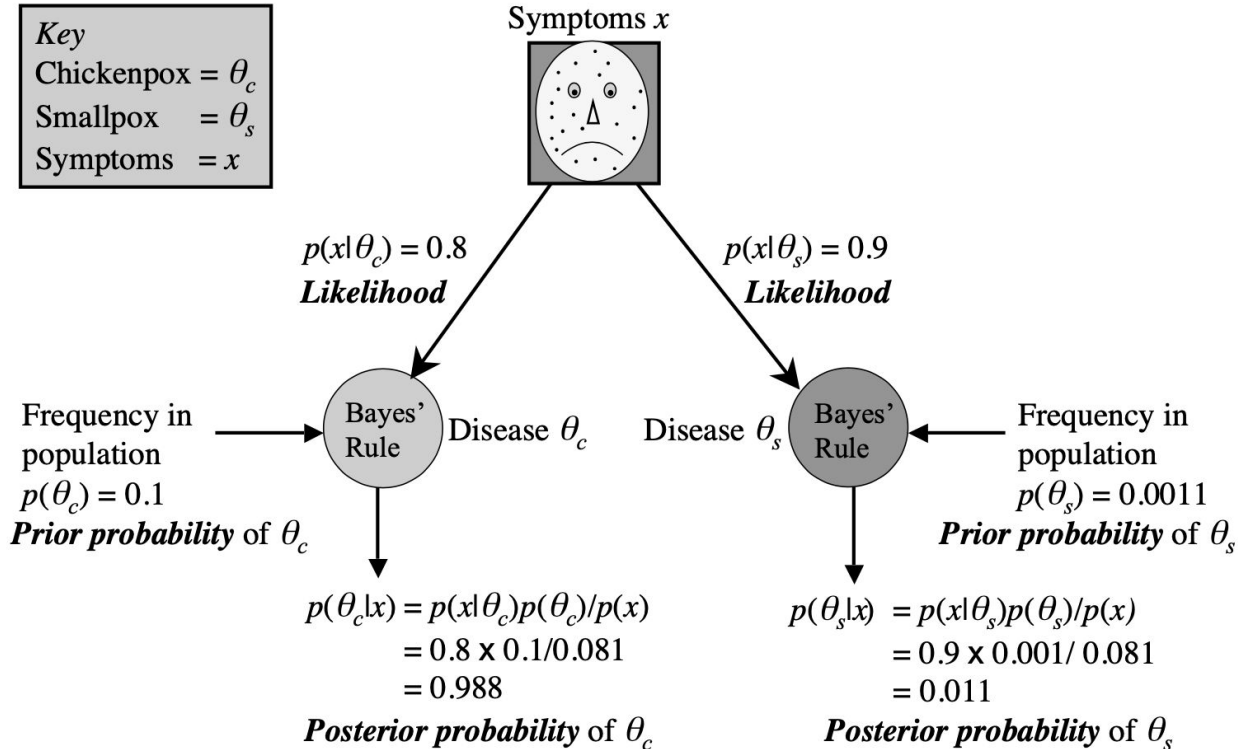


Bayes Rule

$$p(\text{smallpox}|\text{spots}) = \frac{p(\text{spots}|\text{smallpox}) \times p(\text{smallpox})}{p(\text{spots})}.$$

```
# likelihood = prob of spots given smallpox.
pSpotsGSmallpox <- 0.9;
# prior = prob of smallpox.
pSmallpox <- 0.001;
# marginal likelihood = prob of spots.
pSpots <- 0.081;
# find posterior = prob of smallpox given spots.
pSmallpoxGSpots = pSpotsGSmallpox * pSmallpox / pSpots;
#print
s <- sprintf("pSmallpoxGSpots = %.3f", pSmallpoxGSpots)
print(s)
# Output:    pSmallpoxGSpots = 0.011
```

Bayesian Inference



Maximum a Posteriori (MAP) Estimate

Smallpox

$$p(\theta_s|x) = \frac{p(x|\theta_s) \times p(\theta_s)}{p(x)}.$$

Chickenpox

$$p(\theta_c|x) = \frac{p(x|\theta_c) \times p(\theta_c)}{p(x)}.$$

Succinct Notation

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}.$$

$$p(\text{hypothesis}|\text{data}) = \frac{p(\text{data}|\text{hypothesis}) \times p(\text{hypothesis})}{p(\text{data})}$$

Maximum a Posteriori (MAP) Estimate

Smallpox

$$p(\theta_s|x) = \frac{p(x|\theta_s) \times p(\theta_s)}{p(x)}.$$

Chickenpox

$$p(\theta_c|x) = \frac{p(x|\theta_c) \times p(\theta_c)}{p(x)}.$$

$$R_{post} = \frac{p(x|\theta_c)}{p(x|\theta_s)} \times \frac{p(\theta_c)}{p(\theta_s)}.$$

What has cancelled out?

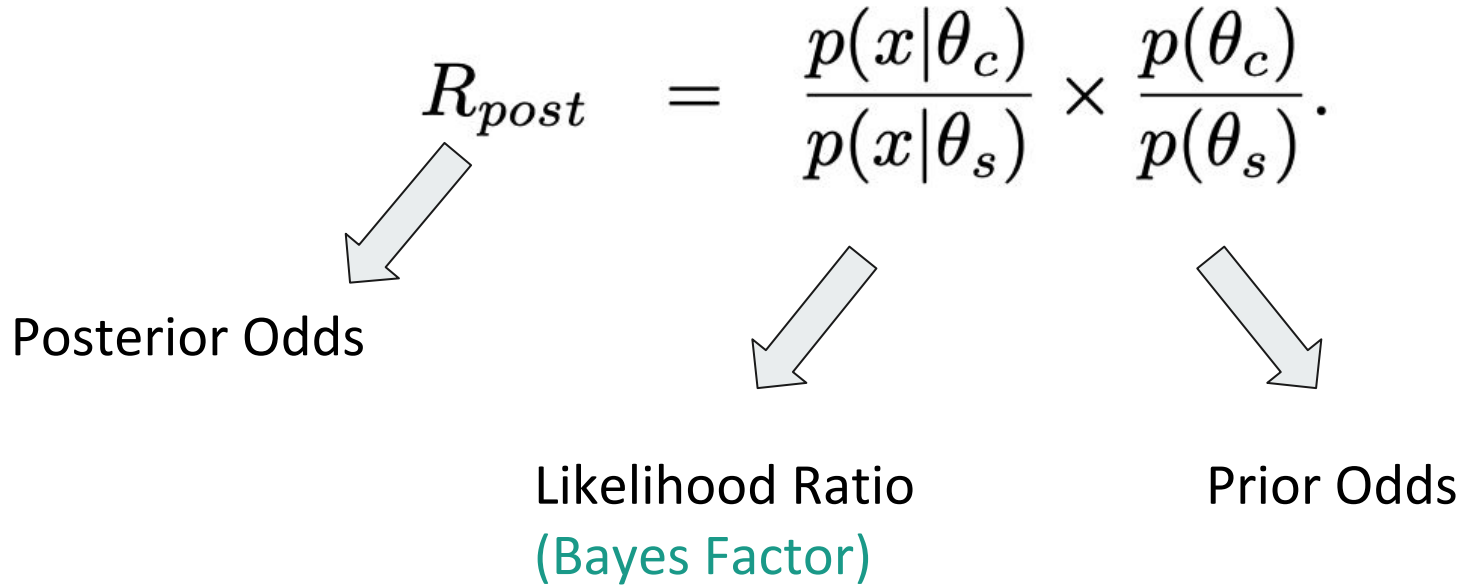
Bayes Factor

$$R_{post} = \frac{p(x|\theta_c)}{p(x|\theta_s)} \times \frac{p(\theta_c)}{p(\theta_s)}.$$

Posterior Odds

Likelihood Ratio
(Bayes Factor)

Prior Odds



The cancelled out term is also called Marginal Likelihood or Evidence

Model Selection

posterior odds = Bayes factor $\times$ prior odds.

$$R_{post} = \frac{0.80}{0.90} \times \frac{0.1}{0.001} = 88.9.$$

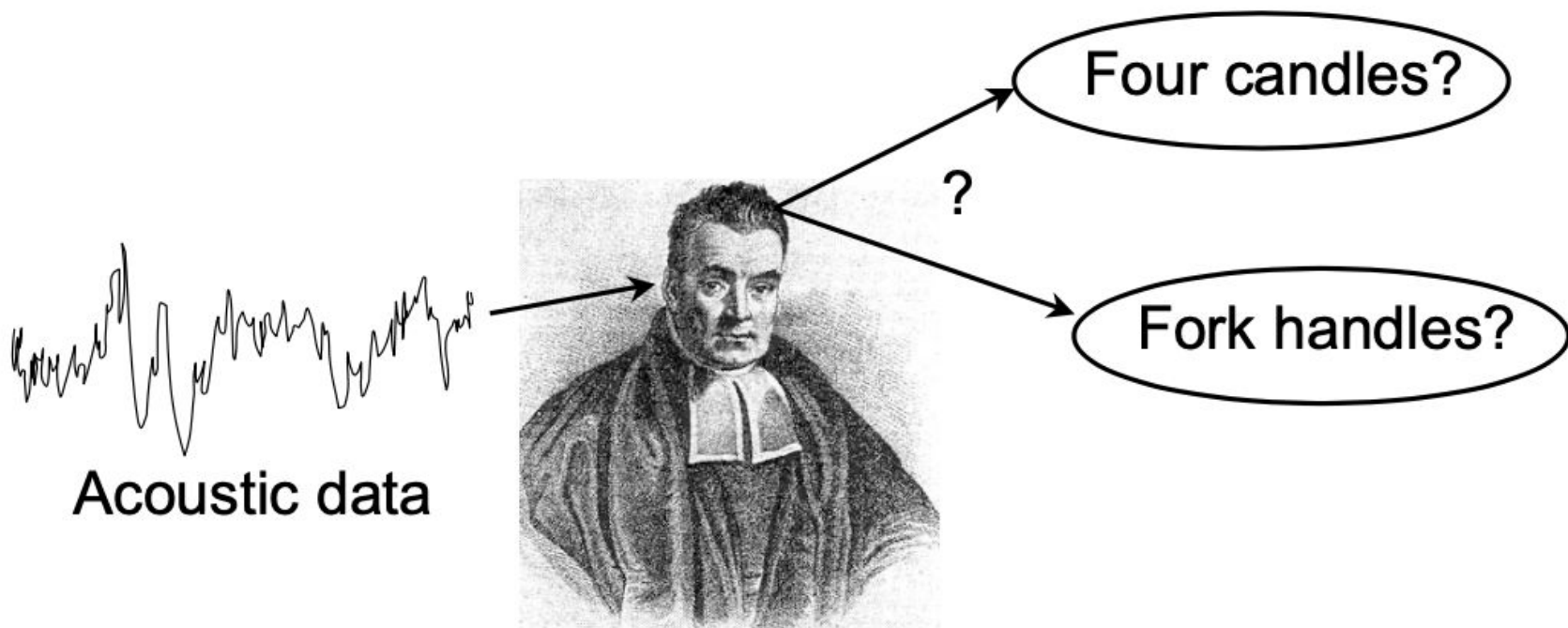
Thumb rule: A posterior odds greater than 3 or less than 1/3 is considered substantial difference between the probabilities of the two models.

When is Bayes Rule Useful?



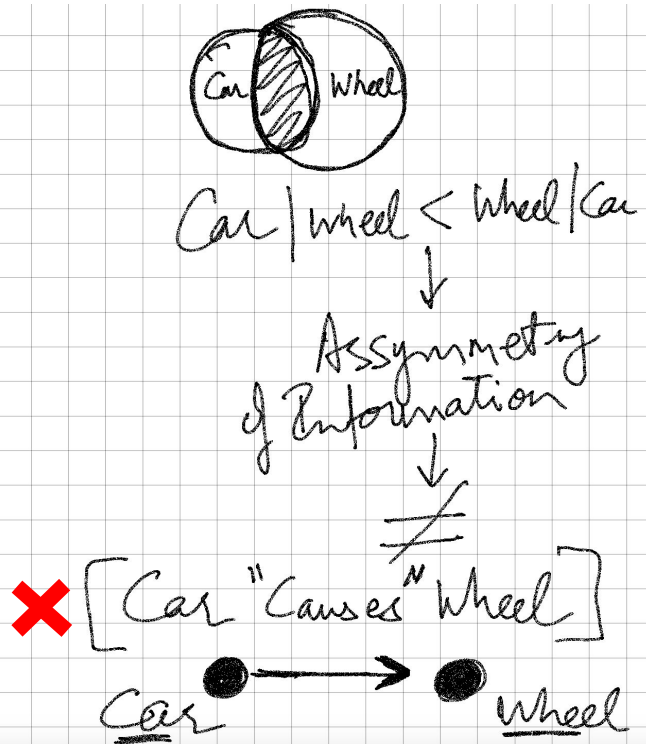
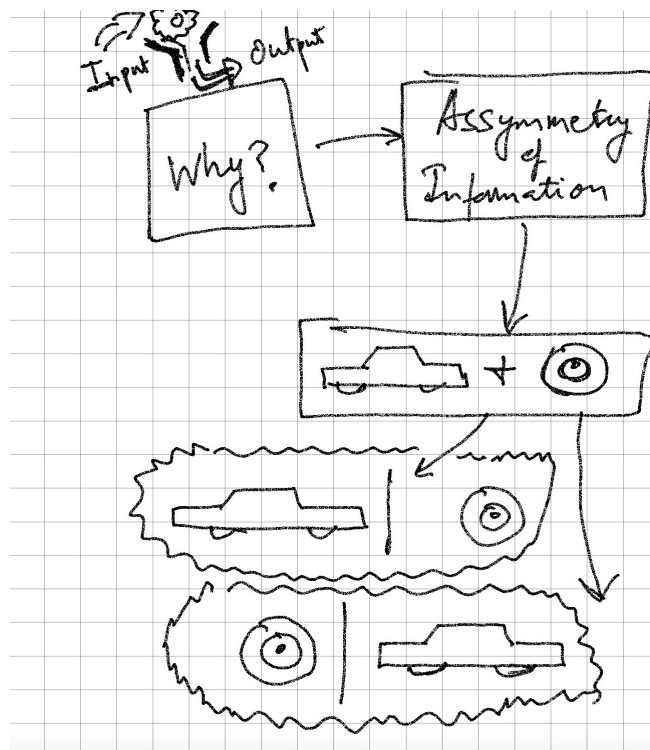
<https://www.youtube.com/watch?v=pV1IP4N9ajg>

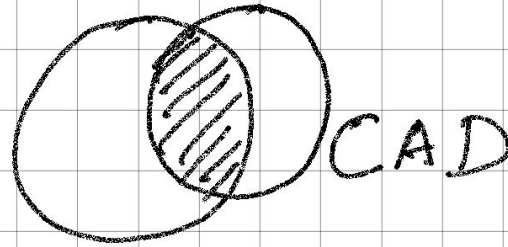
ML with Bayesian Models



BAYESIAN NETWORKS

Bayesian Networks



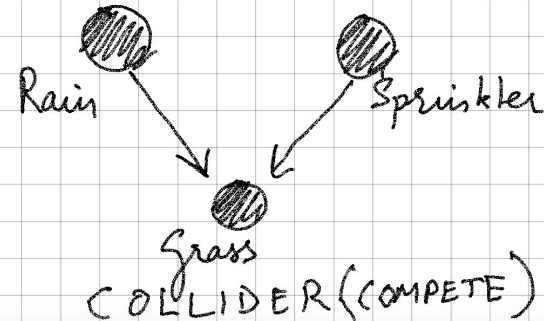
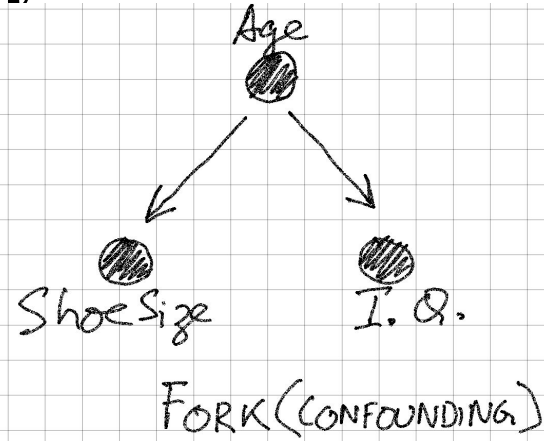
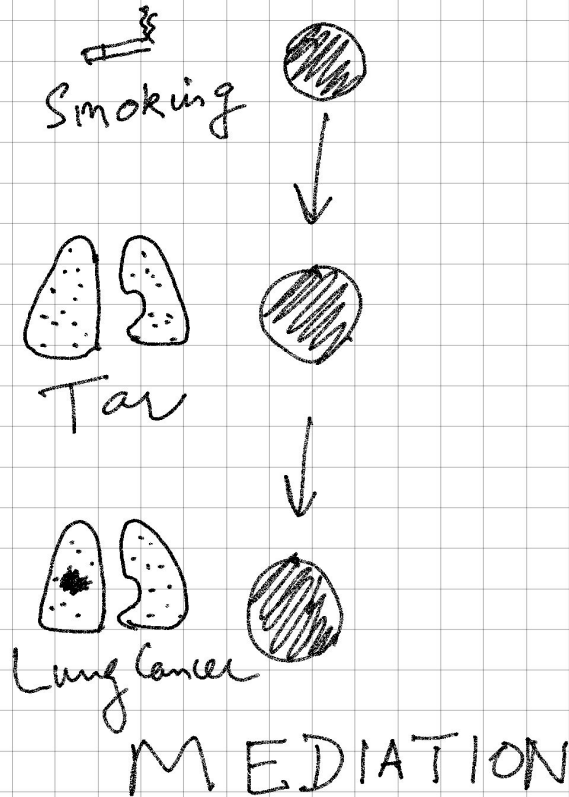


Aspirin

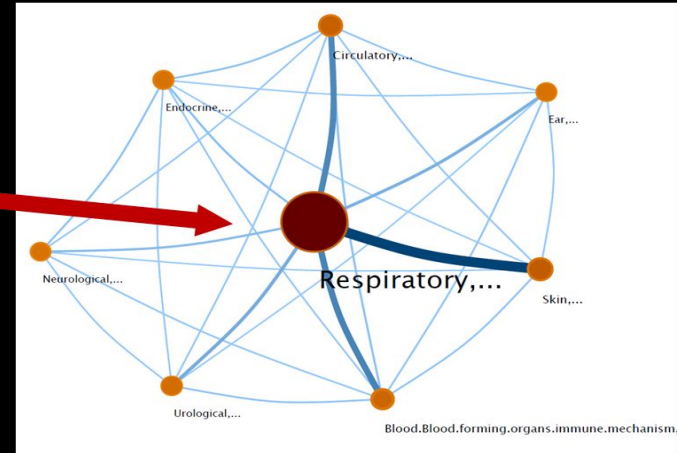
$$P_r(\text{Aspirin} | \text{CAD}) > P_r(\text{CAD} | \text{Aspirin})$$



Building Blocks of Bayesian Nets



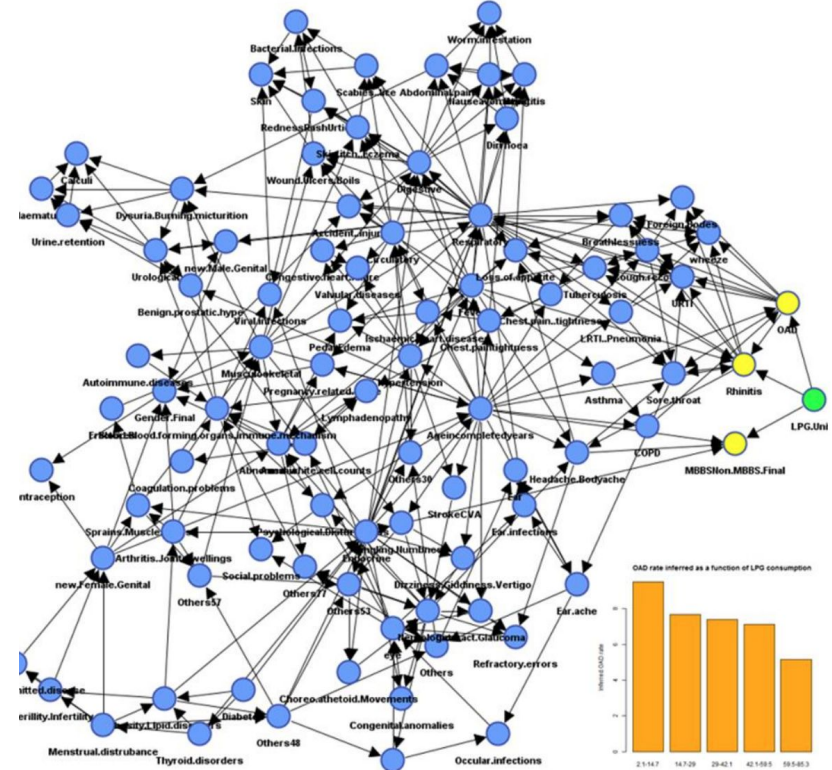
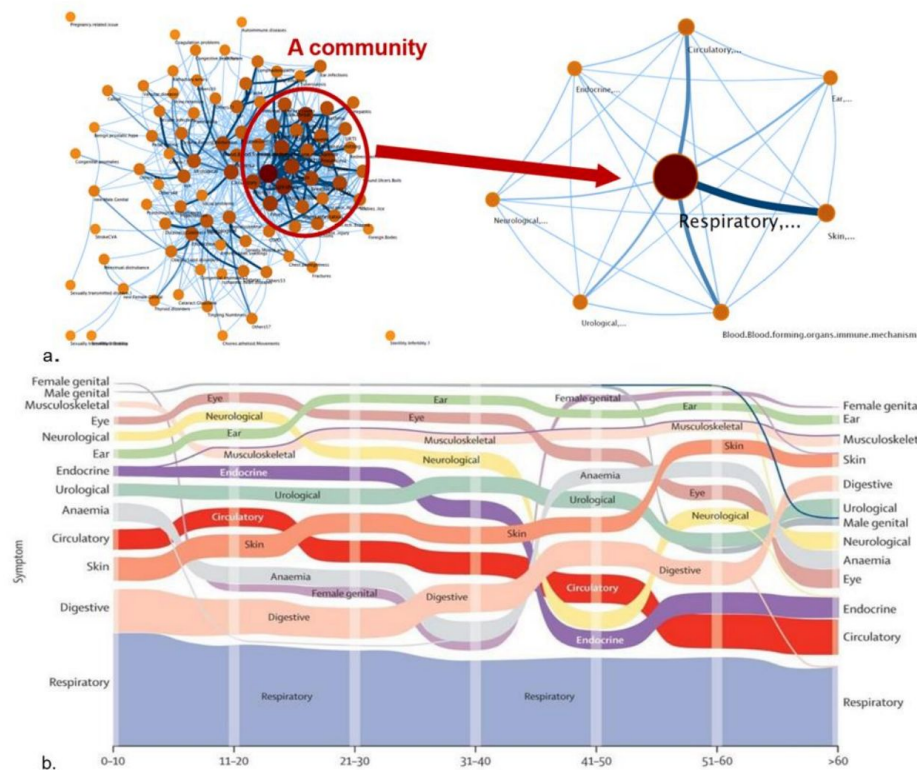
Chest Research Foundation, Pune, India



Networks approach: many diseases happen together, connectivity differs across age
Lancet Global Health, Dec 2015

Sundeeep Salvi et al. Symptoms and medical conditions in 204 912 patients visiting primary health-care practitioners in India: a 1-day point prevalence study (the POSEIDON study), *The Lancet Global Health*, Volume 3, Issue 12, 2015, Pages e776-e784, [https://doi.org/10.1016/S2214-109X\(15\)00152-7](https://doi.org/10.1016/S2214-109X(15)00152-7).

Associations to Decisions



Sundee Salvi et al. Symptoms and medical conditions in 204 912 patients visiting primary health-care practitioners in India: a 1-day point prevalence study (the POSEIDON study), *The Lancet Global Health*, Volume 3, Issue 12, 2015, Pages e776-e784, [https://doi.org/10.1016/S2214-109X\(15\)00152-7](https://doi.org/10.1016/S2214-109X(15)00152-7).

Case Study II: AI for Reducing Health Inequities

- The richest American men live **15 years** longer than the poorest American men¹
- The richest American women live **10 years** longer than the poorest American women¹
- **Healthcare inequities** impose an estimated burden of **\$300 billion per year** in the United States
- Longevity is the **sum-total of influences** on the healthcare
- Hence **longevity-gap** is a complex **socio-demographic** challenge
- Key motivation: **learn policy** for **mitigating** the longevity-gap using explainable AI and release it to public, policymakers

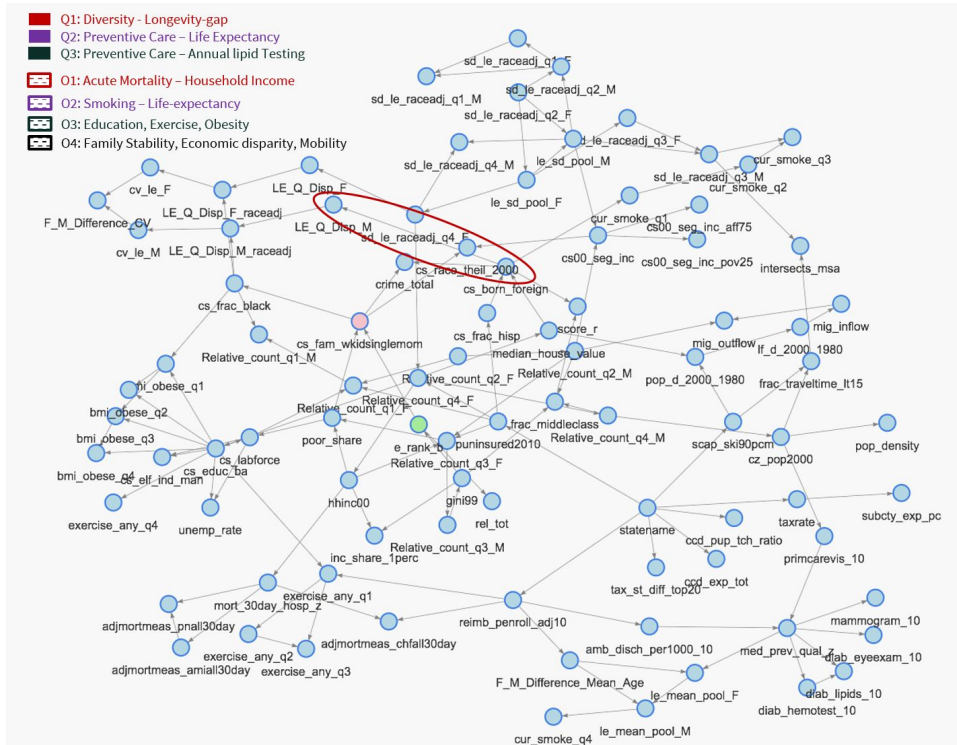
Key Messages

We used data from: **Mortality** (Census), **Healthcare** Indices (Dartmouth Atlas), **Health-behaviors** (CDC, BRFSS), **Education** (K-12 and Post Secondary), **Demographics** (e.g. race, ethnicity, diversity, gender ratio etc), **Socioeconomics** (e.g. Gini index, Poverty rate, Income segregation, Social Mobility), **Social Cohesion**. (e.g. Social Capital Index, Religious adherents), **Labor market conditions and Taxation**. (e.g. unemployment, manufacturing sector).

Key Messages

1. **Social.** Diversity mitigates health inequality in the US. Counties with higher diversity are 38% less likely to have a longevity gap between the rich and the poor.
2. **Preventive Care.** Counties with high quality primary care services (not just visits but investigations) have a 43% increase in the probability of living beyond 85 years in females (corresponding 30% increase for males.)
3. **Clinical.** Acute mortality (30-day Hospital Mortality Index) is 30% less in Counties with household income in the highest segment.
4. **Usual suspects.** Smoking, Education, Exercise as expected to be key influencers of longevity.
5. **Socio-demographic.** Family stability decreases crime rate, increases upward social mobility across economic tiers and indirectly decreases Gini disparity in counties.

Diversity Mitigates Longevity Gap



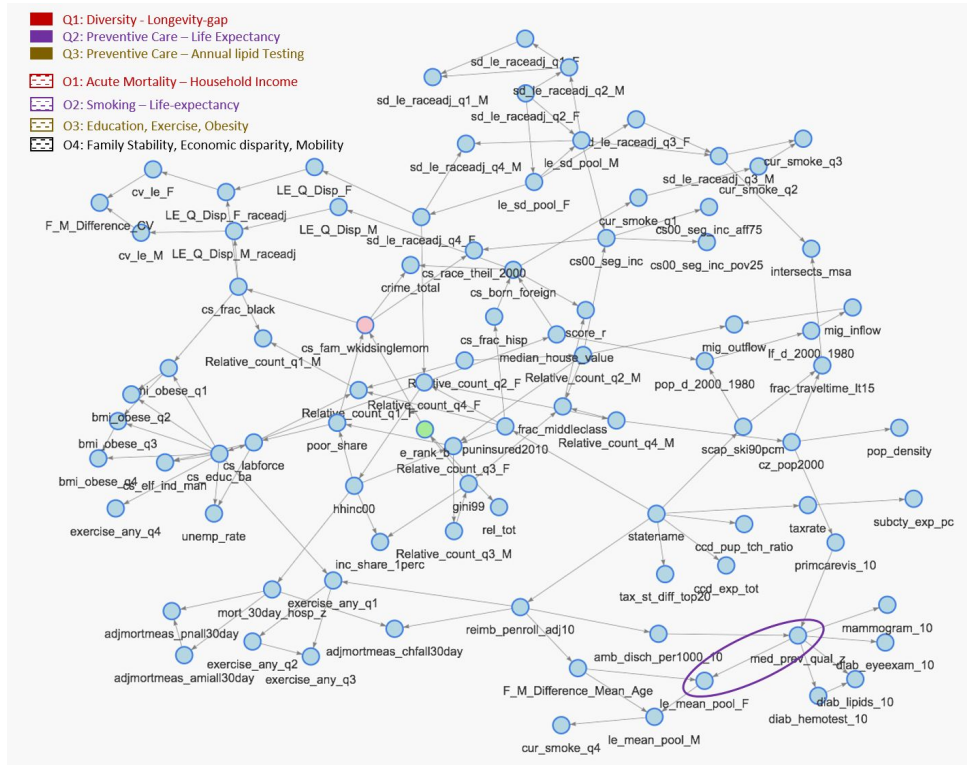
A 1. Diversity

Counties with higher diversity are 38% less likely to have a longevity gap between the rich and the poor.

Likely explanation: Structure indicates that Higher Diversity is Associated with Higher Income, especially in Females, thus improving healthcare services.

Sethi T, S. Maheshwari, A. Mittal, S. Chugh. Learning to Address Health Inequality in the United States with a Bayesian Decision Network. Proceedings of the AAAI Conference on Artificial Intelligence 33, 710-717. DOI: <https://doi.org/10.1609/aaai.v33i01.3301710> c

The Impact of Primary Care



A 2. Quality of Preventive Care

Counties with high quality primary care services have a 43% increase in the probability of living beyond 85 years in females (corresponding 30% increase for males.)

Likely explanation: Self Evident, but previously unquantified in a Joint Model

Sethi T, S. Maheshwari, A. Mittal, S. Chugh. Learning to Address Health Inequality in the United States with a Bayesian Decision Network. Proceedings of the AAAI Conference on Artificial Intelligence 33, 710-717. DOI: <https://doi.org/10.1609/aaai.v33i01.3301710> c

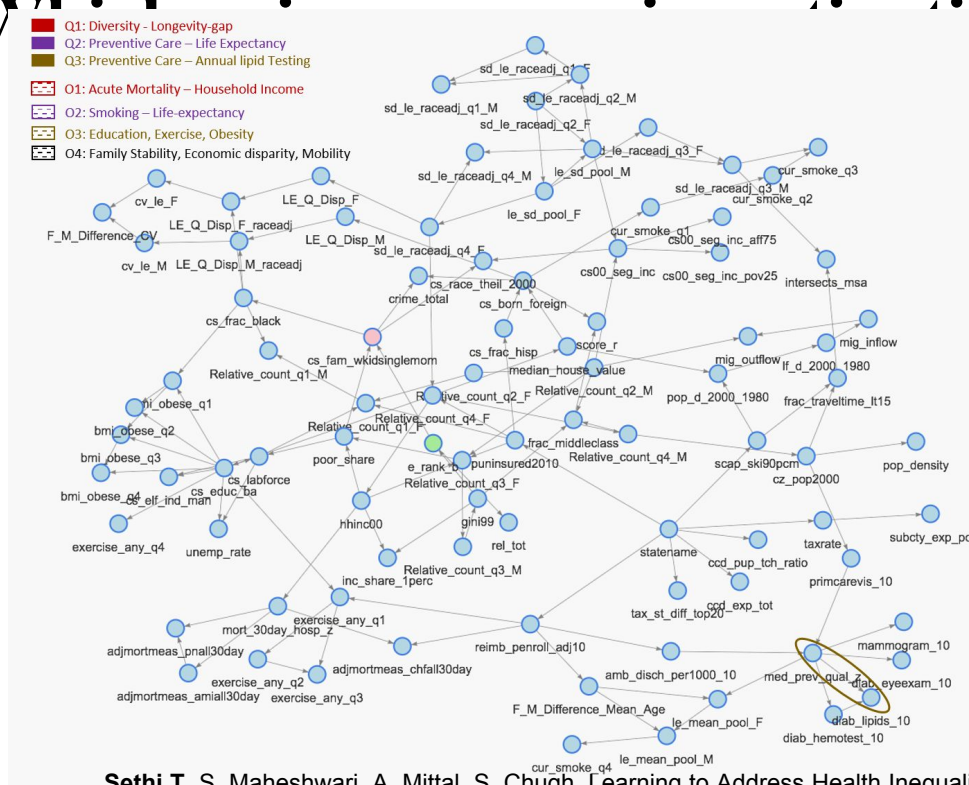
What has most impact?

A 3. Annual Lipid Testing

esp. in the diabetic population

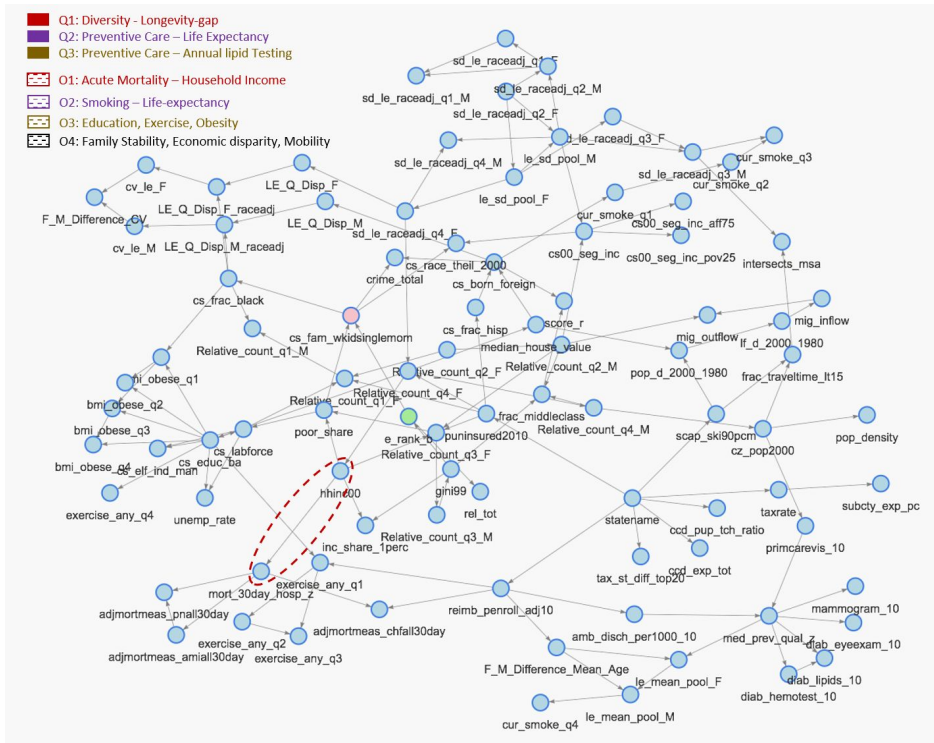
diab_eyeexam_10	mammogram_10	diab_lipids_10	payoff
[70.2,85.6]	[68.2,86.1]	[79.3,92.9]	0.52
[70.2,85.6]	[68.2,86.1]	[65.6,79.3]	0.48
[62.2,70.2]	[68.2,86.1]	[79.3,92.9]	0.44
[70.2,85.6]	[59.5,68.2]	[79.3,92.9]	0.40
[62.2,70.2]	[68.2,86.1]	[65.6,79.3]	0.26
[70.2,85.6]	[59.5,68.2]	[65.6,79.3]	0.22
[42.4,62.2]	[68.2,86.1]	[79.3,92.9]	0.20
[62.2,70.2]	[59.5,68.2]	[79.3,92.9]	0.12
[70.2,85.6]	[31.1,59.5]	[79.3,92.9]	0.11
[42.4,62.2]	[68.2,86.1]	[65.6,79.3]	0.08

Likely explanation:
 Diabetics are at highest risk of cardiovascular mortality



Sethi T, S. Maheshwari, A. Mittal, S. Chugh. Learning to Address Health Inequality in the United States with a Bayesian Decision Network. Proceedings of the AAAI Conference on Artificial Intelligence 33, 710-717. DOI: <https://doi.org/10.1609/aaai.v33i01.3301710> c

Income and Acute Mortality

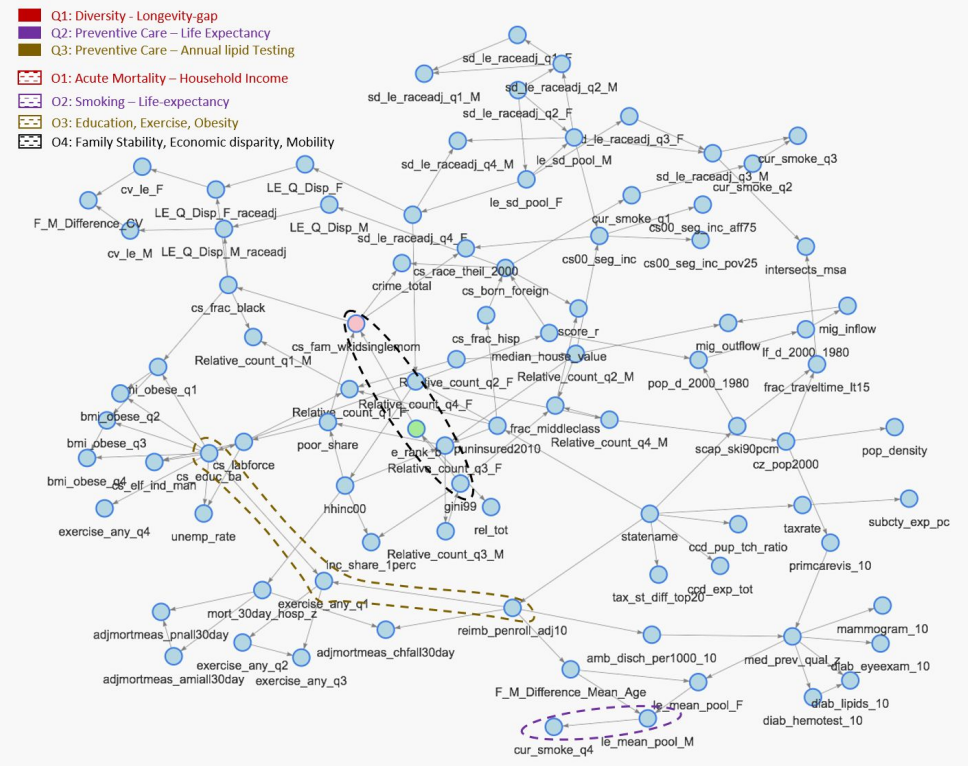


O 1.

ACUTE MORTALITY (30-DAY HOSPITAL MORTALITY INDEX > 0.92) IS 30% LESS IN COUNTIES WITH HOUSEHOLD INCOME IN THE HIGHEST SEGMENT.

HIGHEST CONTRIBUTOR TO
ACUTE MORTALITY IN LOWER
INCOME HOUSEHOLDS IS
PNEUMONIA

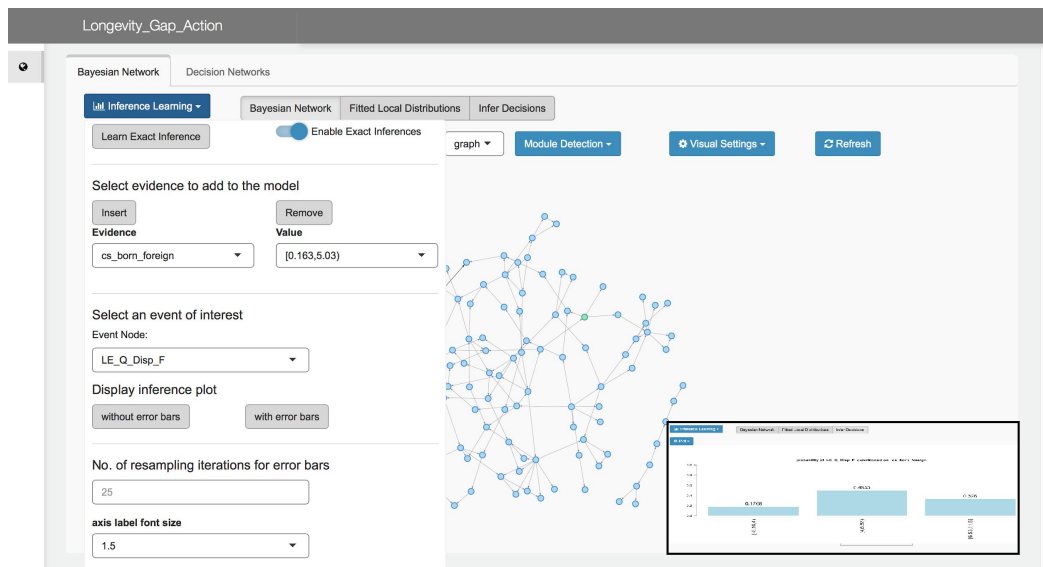
Smoking, Exercise, Health Behaviors



USUAL SUSPECTS (O2-O4)

- HIGH LIFE-EXPECTANCY IN MALES (81.9 - 85 YEARS) MAKES IT 33% MORE LIKELY FOR **SMOKING** TO BE IN THE LOWEST STRATUM IN THE COUNTY.
- COUNTIES WITH HIGH PROPORTION OF EXERCISE HAVE 19% LESS HOSPITALIZATION RATES
- COUNTIES WITH LOWER FAMILY STABILITY ARE 40% MORE LIKELY TO HAVE LOWER SOCIAL MOBILITY

Deploy your XAI models as Web applications



https://github.com/SAFE-ICU/Longevity_Gap_Action

Key Messages

1. **Social.** Diversity mitigates health inequality in the US. Counties with higher diversity are 38% less likely to have a longevity gap between the rich and the poor.
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3. **Clinical.** Acute mortality (30-day Hospital Mortality Index) is 30% less in Counties with household income in the highest segment.
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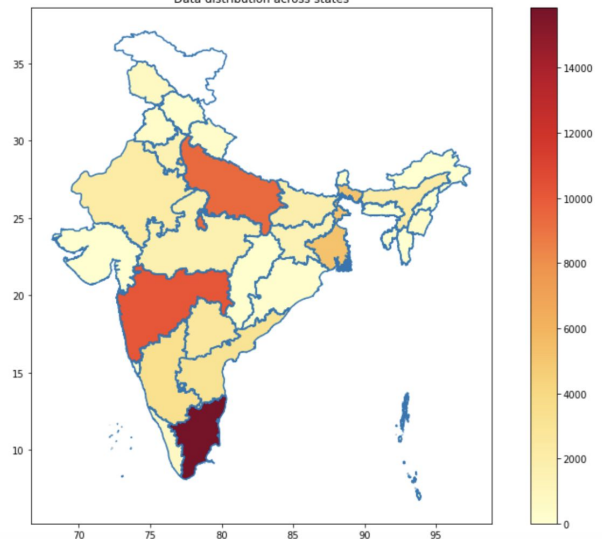
Tavpritesh Sethi, Anant Mittal, Shubham Maheshwari, Samarth Chugh. *Learning to Address Health Inequality in the United States with a Bayesian Decision Network.* <https://arxiv.org/abs/1809.09215> Accepted for publication in the Thirty-third AAAI conference in Artificial Intelligence, AAAI-2019

Emerging trends in antimicrobial resistance in bloodstream infections: multicentric longitudinal study in India (2017–2022)

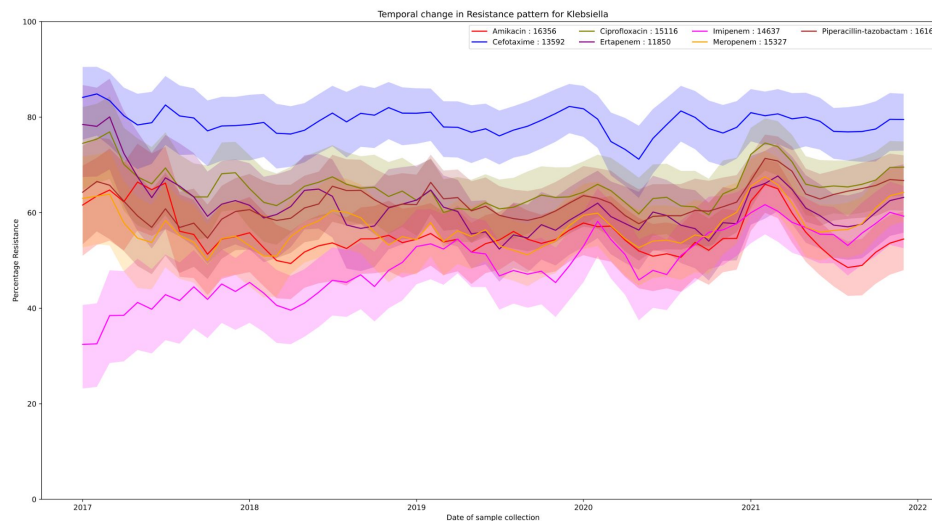
[Jasmine Kaur](#)^{a,b,c} · [Harpreet Singh](#)^c · [Tavprites Sethi](#)^{a,b}  



Data distribution across states



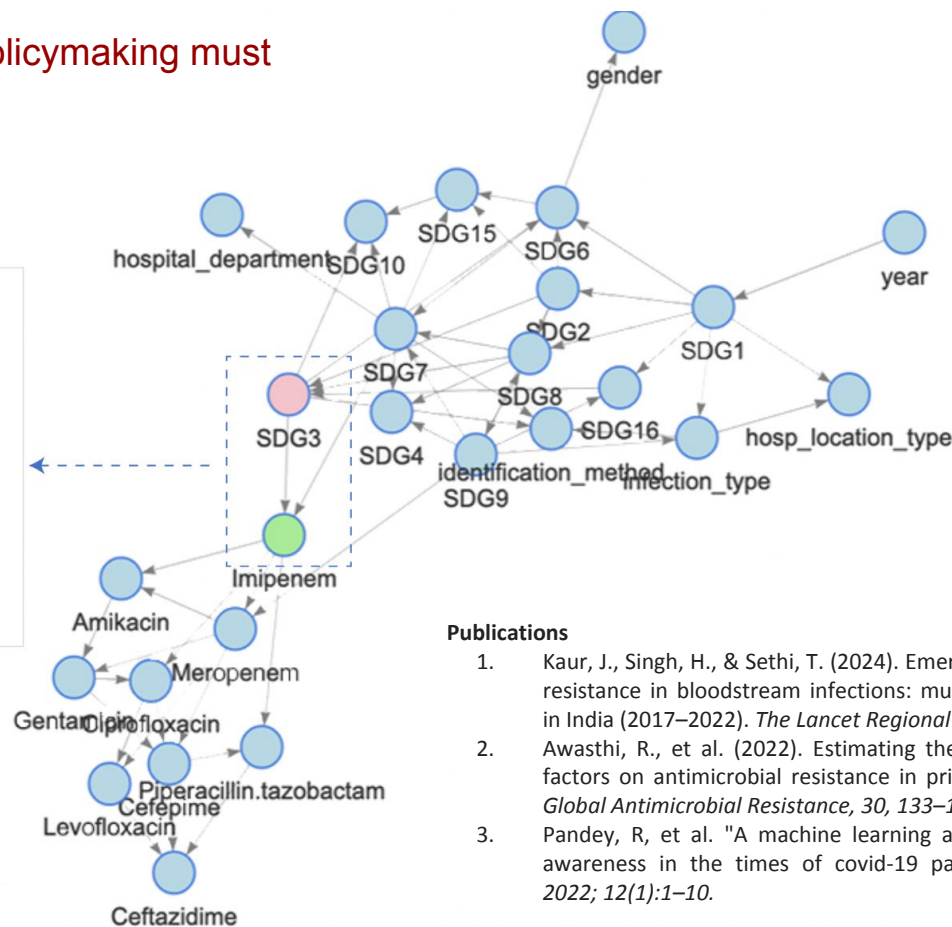
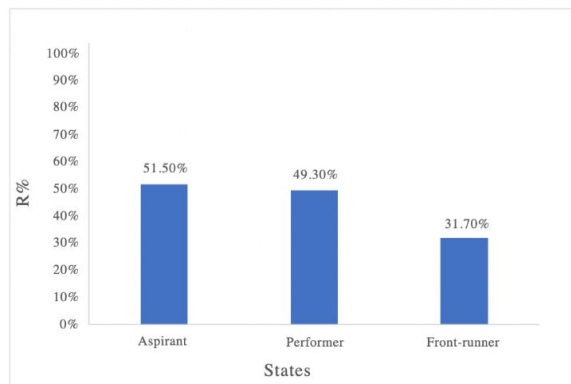
Klebsiella Sepsis



0.25% monthly average for increase in Imipenem resistance

AI for Understanding Impact on SDG Achievement

Key Takeaway: Evidence-based policymaking must leverage AI for tracking progress.



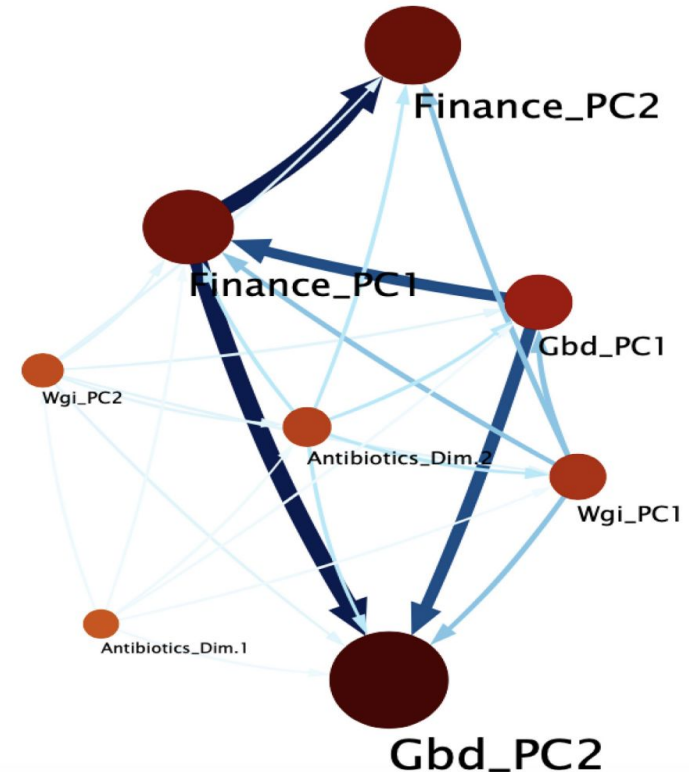
Publications

1. Kaur, J., Singh, H., & Sethi, T. (2024). Emerging trends in antimicrobial resistance in bloodstream infections: multicentric longitudinal study in India (2017–2022). *The Lancet Regional Health-Southeast Asia*, 26.
2. Awasthi, R., et al. (2022). Estimating the impact of health systems factors on antimicrobial resistance in priority pathogens. *Journal of Global Antimicrobial Resistance*, 30, 133–142.
3. Pandey, R, et al. "A machine learning application for raising wash awareness in the times of covid-19 pandemic". *Scientific reports* 2022; 12(1):1–10.

Systems Indicators and AMR

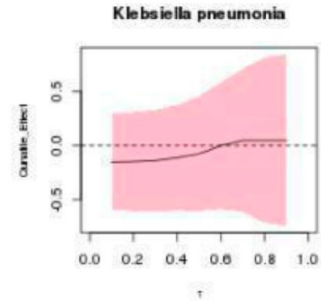
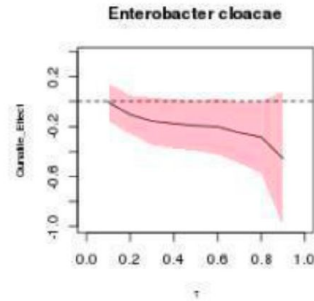
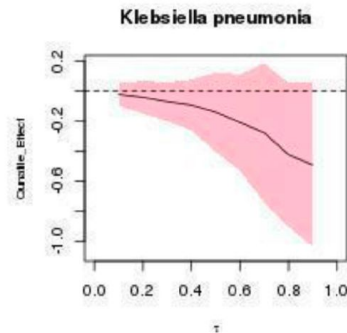
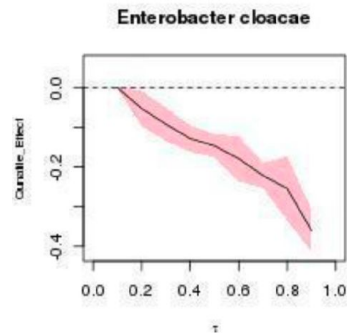
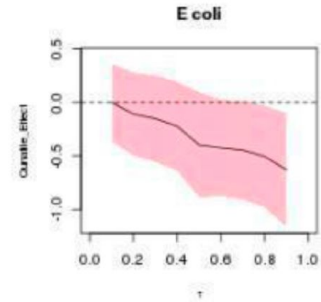
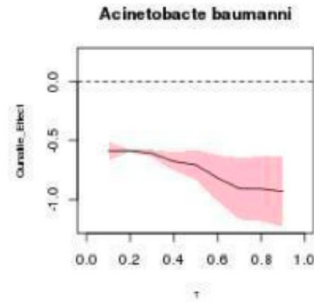
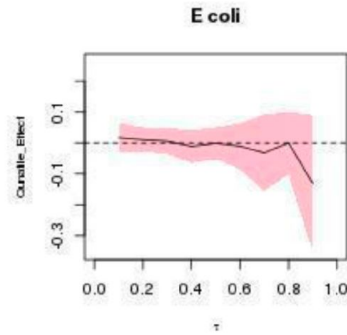
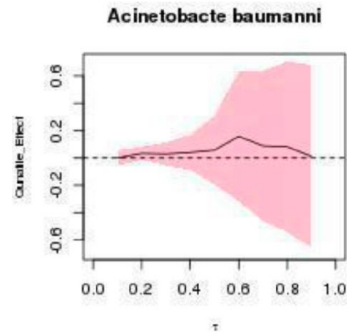
Age and temporal distribution of data

	Women	Men	Overall
Sex	No. (%)	No. (%)	No. (%)
	278 128 (43.88%)	348 136 (54.93%)	633 820
Age group			
0 to 2 years	15 329 (42.83%)	20 012 (55.91%)	35 793
13 to 18 years	6348 (47.06%)	7043 (52.21%)	13 490
19 to 64 years	130 162 (44.04%)	163 201 (55.22%)	295 537
3 to 12 years	12 613 (46.88%)	14 052 (52.22%)	26 907
65 to 84 years	86 561 (41.63%)	119 801 (57.62%)	207 922
85 and over	23 406 (54.29%)	19 364 (44.91%)	43 114
Year			
2004	9433 (46.93%)	10 655 (53.01%)	20 101
2005	10 473 (48.04%)	11 274 (51.71%)	21 801
2006	13 967 (46.65%)	15 876 (53.03%)	29 940
2007	18 264 (45.70%)	21 409 (53.57%)	39 964
2008	16 303 (44.33%)	19 925 (54.18%)	36 773
2009	18 575 (44.55%)	22 514 (54.00%)	41 692
2010	14 476 (44.59%)	17 341 (53.42%)	32 462
2011	11 622 (44.66%)	13 931 (53.53%)	26 023
2012	22 432 (42.16%)	29 477 (55.40%)	53 206
2013	29 962 (42.73%)	38 850 (55.40%)	70 125
2014	30 492 (43.23%)	39 482 (55.98%)	70 529
2015	27 992 (43.21%)	36 104 (55.73%)	64 785
2016	29 744 (42.74%)	39 290 (56.45%)	69 598
2017	24 393 (42.93%)	32 008 (56.33%)	56 821



Awasthi R, Rakholia V, Agrawal S, Dhingra LS, Nagori A, Kaur H, Sethi T. Estimating the impact of health systems factors on antimicrobial resistance in priority pathogens. *J Glob Antimicrob Resist.* 2022 Sep;30:133-142.

Impact Calculation: Counterfactual Analysis



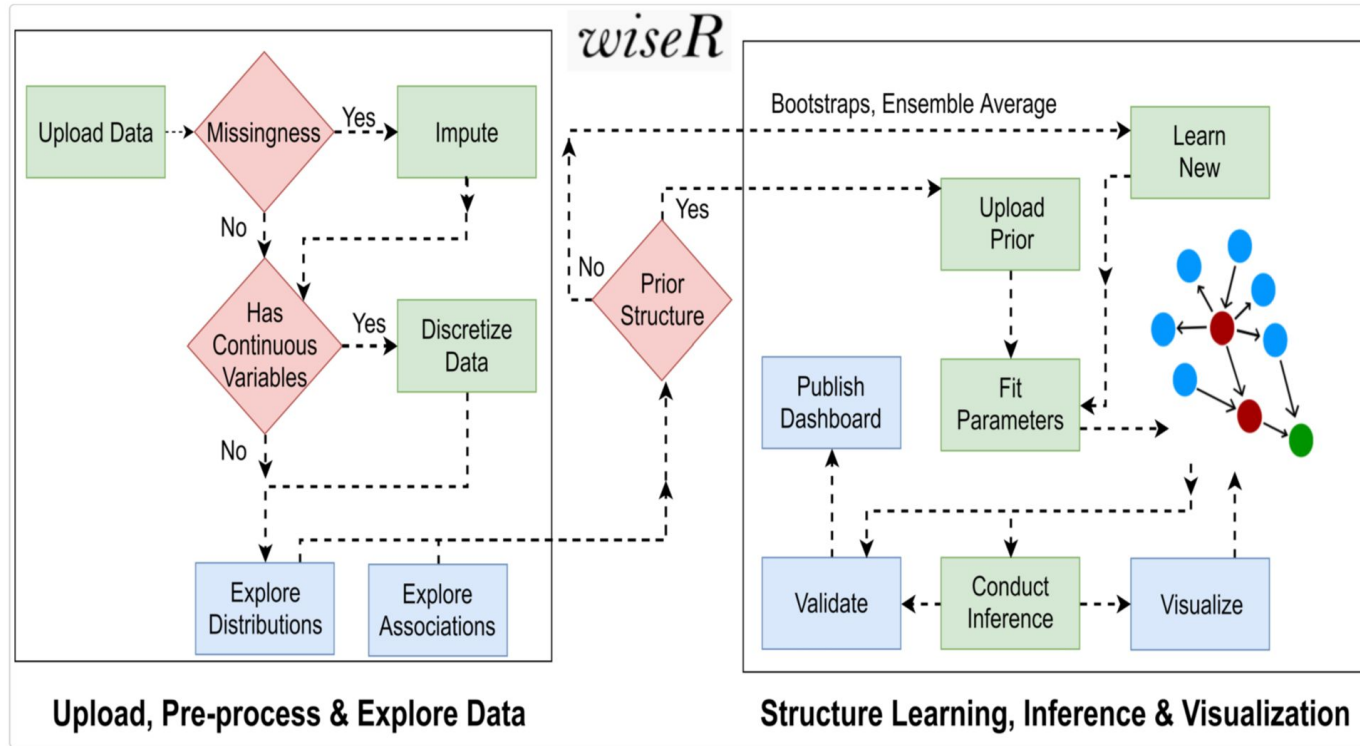
Ceftriaxone High Income

Ceftriaxone Middle Income

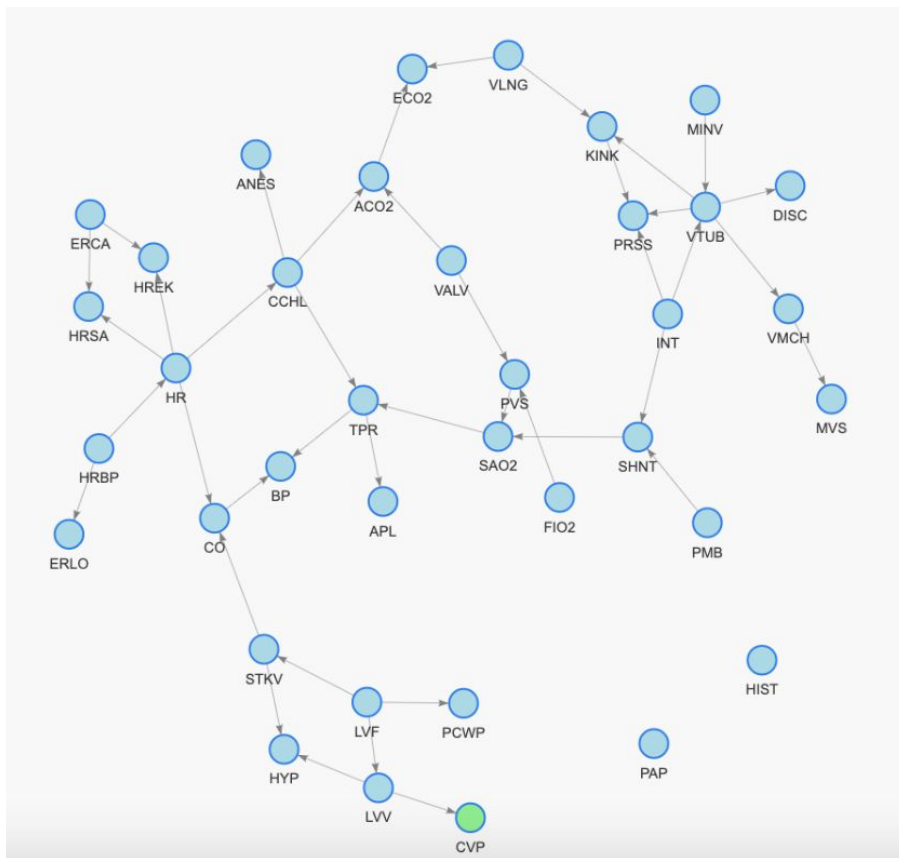
Key Steps in Building a BN Model

- Learn Structure
 - Score Based
 - Constraint Based
- Validate Structure
 - Bootstrapping
 - Domain Based Sanitization
- Conduct Inference
 - Exact Inference
 - Approximate Inference

wiseR: Bayesian Artificial Intelligence



Learning Structure (ICU Alarms Example)



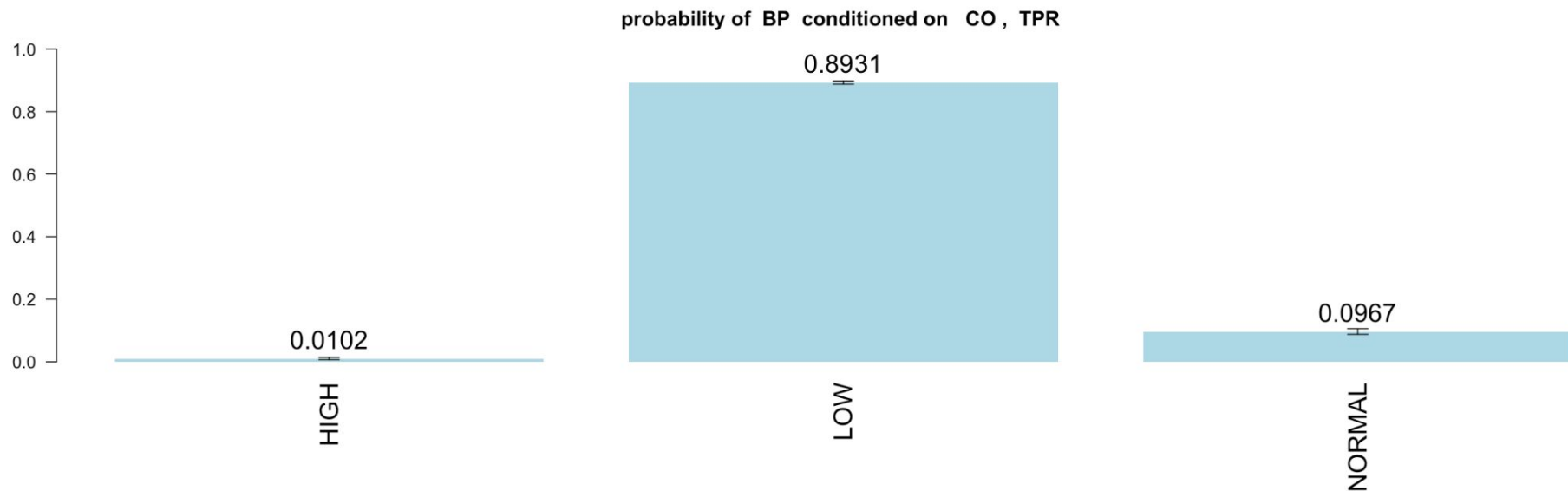
<https://github.com/tavlab-iiitd/wiseR>

Example Set up

We learn the Network structure using the following parameters:

- Algorithm: Hill Climbing
- Network Score: Akaike Information Criterion
- Parameter Fitting: Bayesian Parameter Estimation
- Blacklisting: No
- Bootstrap Model: Yes
- No. of Bootstrap: 101
- Edge Strength: >0.51
- Direction Strength: >0.51
- Sample Portion: 1

Conducting Inference



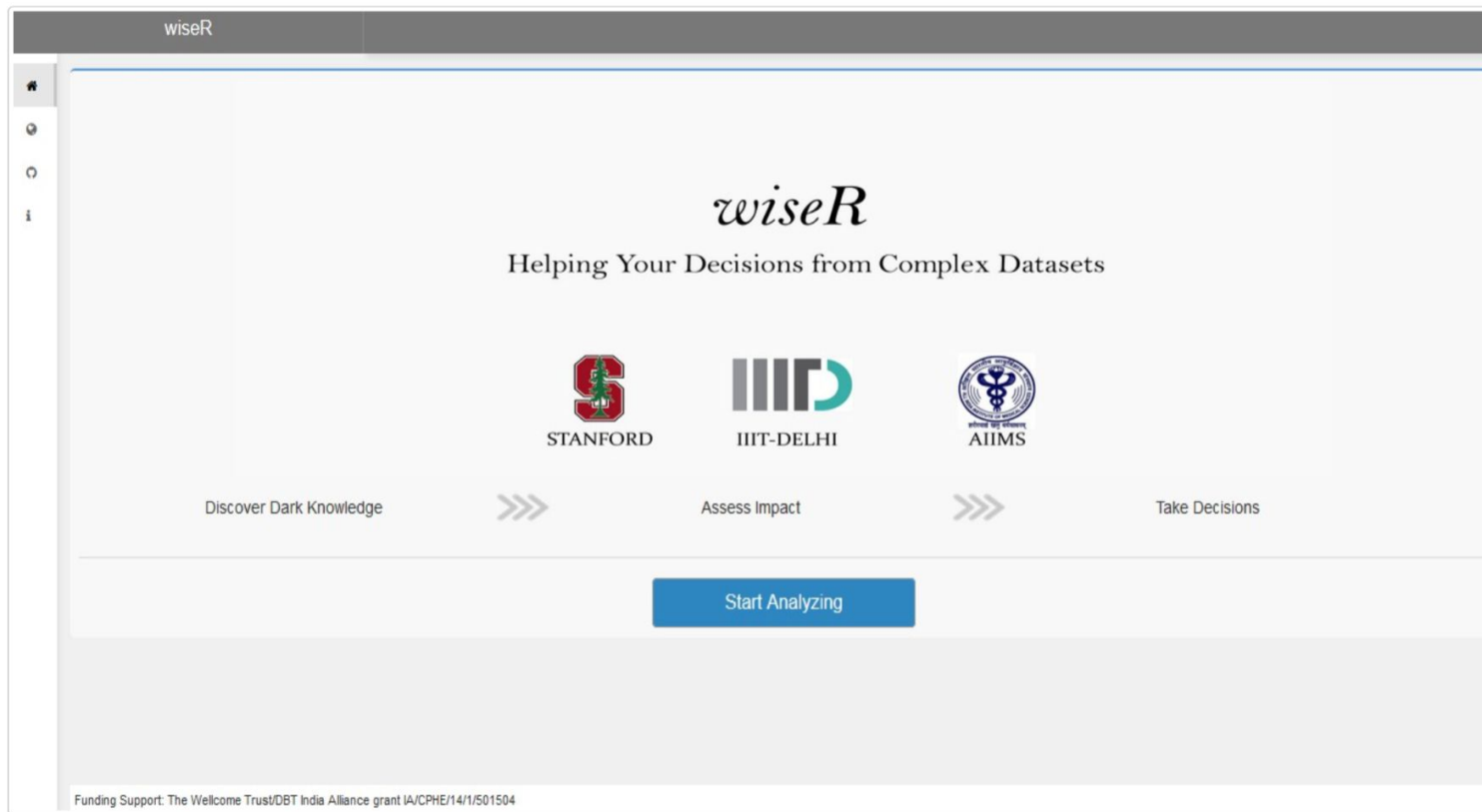
Probabilistic Reasoning

Examples: Anaphylaxis and Total Peripheral Resistance (APL, TPR). Baseline rate of APL: 1%, knowing that the patient had low TPR changes our “probabilistic belief” to 3%

Motifs:

1. Mediation effect: $CCHL \rightarrow MINV \dots \rightarrow ACO_2$
2. Confounding effect
3. Inter-causal reasoning: $CO \rightarrow BP \leftarrow TPR$
 - $CO \leftarrow BP \rightarrow HR$

wiseR: Bayesian Artificial Intelligence



The image shows a screenshot of the wiseR web application interface. At the top, there is a dark grey header bar with the text "wiseR" in white. Below the header, on the left side, is a vertical sidebar with a home icon and three circular icons. The main content area has a light grey background. In the center, the word "wiseR" is written in a large, elegant, italicized serif font. Below it, the tagline "Helping Your Decisions from Complex Datasets" is displayed in a smaller, sans-serif font. Further down, three logos are arranged horizontally: the Stanford University logo (a red block 'S' with a green redwood tree in front of it), the IIT-Delhi logo (the letters "IITD" in a stylized grey and teal font), and the All India Institute of Medical Sciences (AIIMS) logo (a circular emblem with a caduceus in the center and text in Hindi and English). Below these logos, a horizontal flow of three steps is shown: "Discover Dark Knowledge" followed by a double right-pointing arrow, "Assess Impact" followed by another double right-pointing arrow, and "Take Decisions". At the bottom center of the main content area is a prominent blue rectangular button with the white text "Start Analyzing". At the very bottom of the page, a small line of text reads: "Funding Support: The Wellcome Trust/DBT India Alliance grant IA/CPHE/14/1/501504".

wiseR

wiseR

Helping Your Decisions from Complex Datasets

STANFORD

IIT-D

AIIMS

Discover Dark Knowledge >>> Assess Impact >>> Take Decisions

Start Analyzing

Funding Support: The Wellcome Trust/DBT India Alliance grant IA/CPHE/14/1/501504

Practical Session

The screenshot displays the wiseR web application interface. At the top, a browser address bar shows the URL 127.0.0.1:3493. The application header features the 'wiseR' logo on the left and a star icon on the right. A vertical sidebar on the left contains four icons: a home icon, a globe icon, a refresh icon, and an information icon. The main content area has a horizontal navigation bar with the following tabs: 'App Settings' (which is the active tab), 'Data', 'Association Network', 'Bayesian Network', 'Decision Network', 'Publish your dashboard', and 'App Tutorial'. Below the navigation bar, the 'App Settings' section contains a toggle switch for 'Enable Parallel Computing' (currently turned off) and a dropdown menu for 'Number of clusters' (currently set to 1).

wiseR

App Settings Data Association Network Bayesian Network Decision Network Publish your dashboard App Tutorial

Enable Parallel Computing

Number of clusters

1

Loading Data

wiseR

🏠

🌐

🔄

📄

App Settings

Data

Association Network

Bayesian Network

Decision Network

Publish your dashboard

App Tutorial

Dataset

Explore

upload

Pre-Process

Download

Choose default dataset

Alarm

load

Data Format:

.CSV

variables as Factor:

☒

File Input:

Browse...

No file selected

Search:

	HRBP	HREK	HRSA	PAP	SAO2	FIO2	PRSS	ECO2	MINV	MVS	F	
	HIGH	HIGH	HIGH	NORMAL	NORMAL	LOW	HIGH	ZERO	HIGH	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	HIGH	ZERO	ZERO	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	NORMAL	ZERO	ZERO	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	NORMAL	NORMAL	HIGH	ZERO	ZERO	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	LOW	ZERO	ZERO	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	HIGH	HIGH	ZERO	NORMAL	FA	
	HIGH	HIGH	HIGH	NORMAL	LOW	LOW	LOW	ZERO	ZERO	NORMAL	FA	
8	NORMAL	NORMAL	FALSE	NORMAL	LOW	LOW	HIGH	NORMAL	NORMAL	LOW	LOW	FA
9	NORMAL	LOW	FALSE	LOW	LOW	HIGH	HIGH	HIGH	HIGH	NORMAL	LOW	FA
10	NORMAL	NORMAL	FALSE	NORMAL	NORMAL	HIGH	HIGH	HIGH	LOW	LOW	NORMAL	FA

Showing 1 to 10 of 20,000 entries

Previous

1

2

3

4

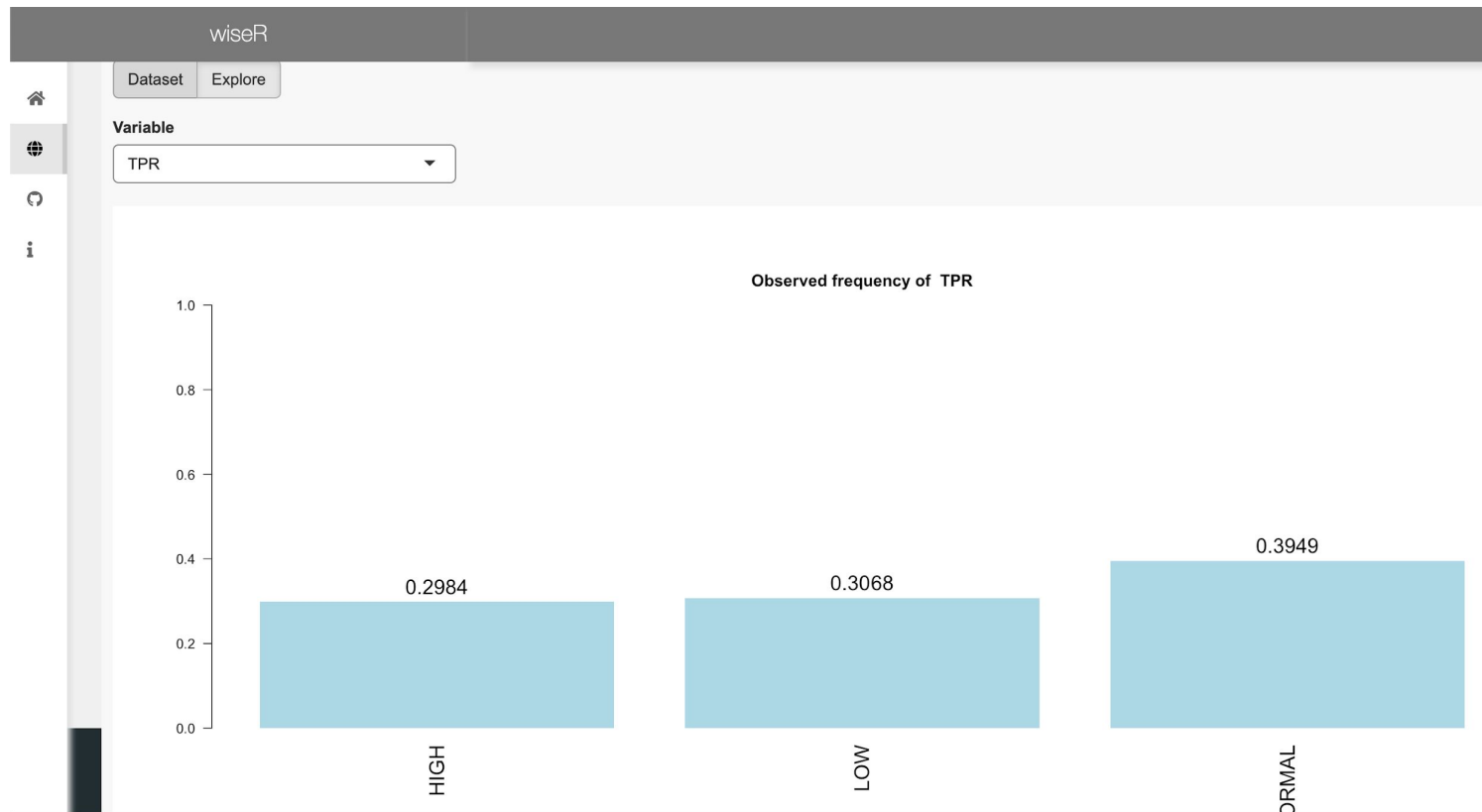
5

...

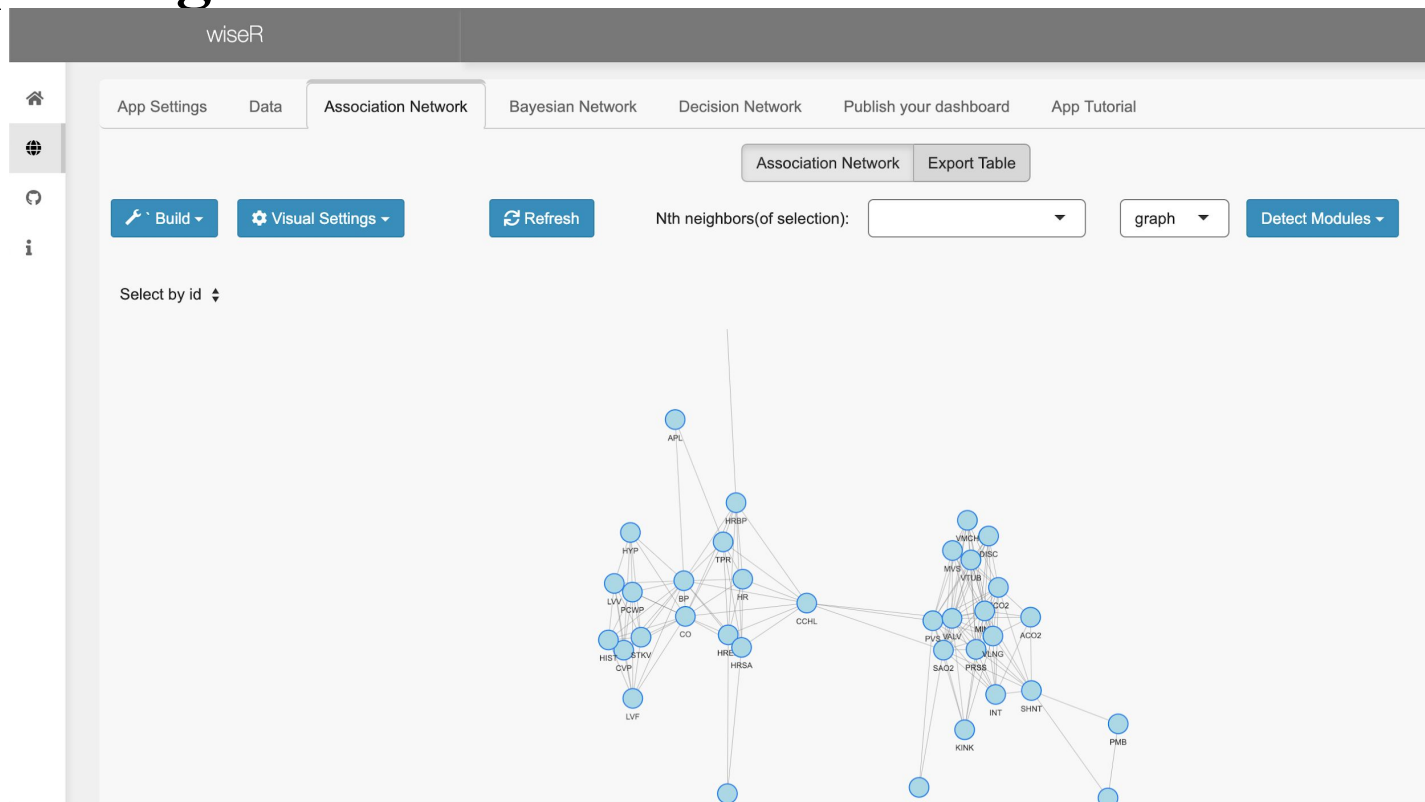
2000

Next

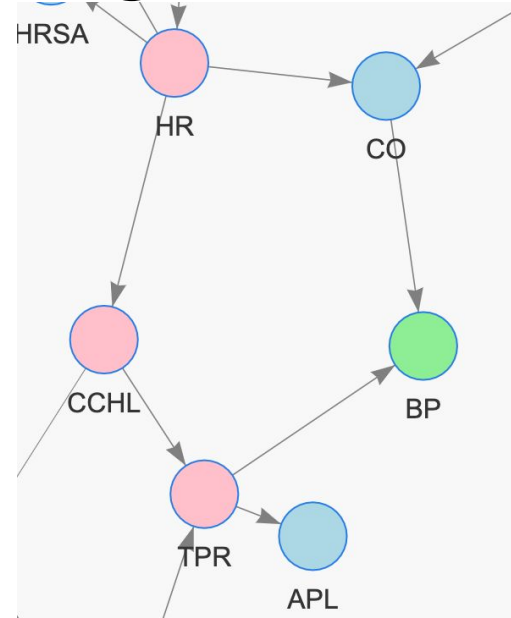
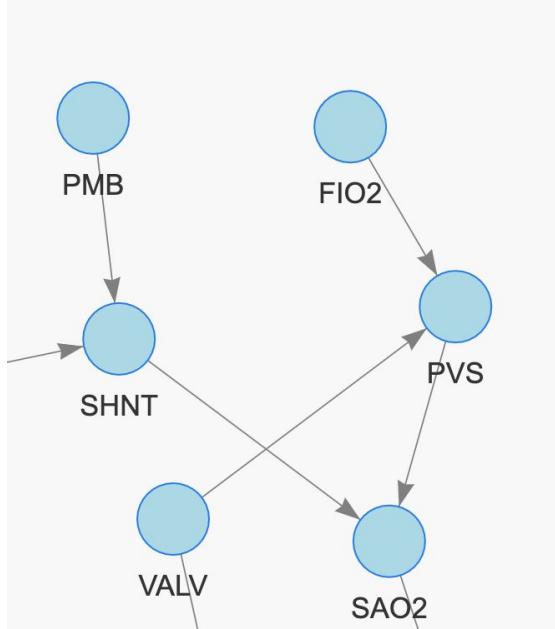
Exploration of univariate distributions



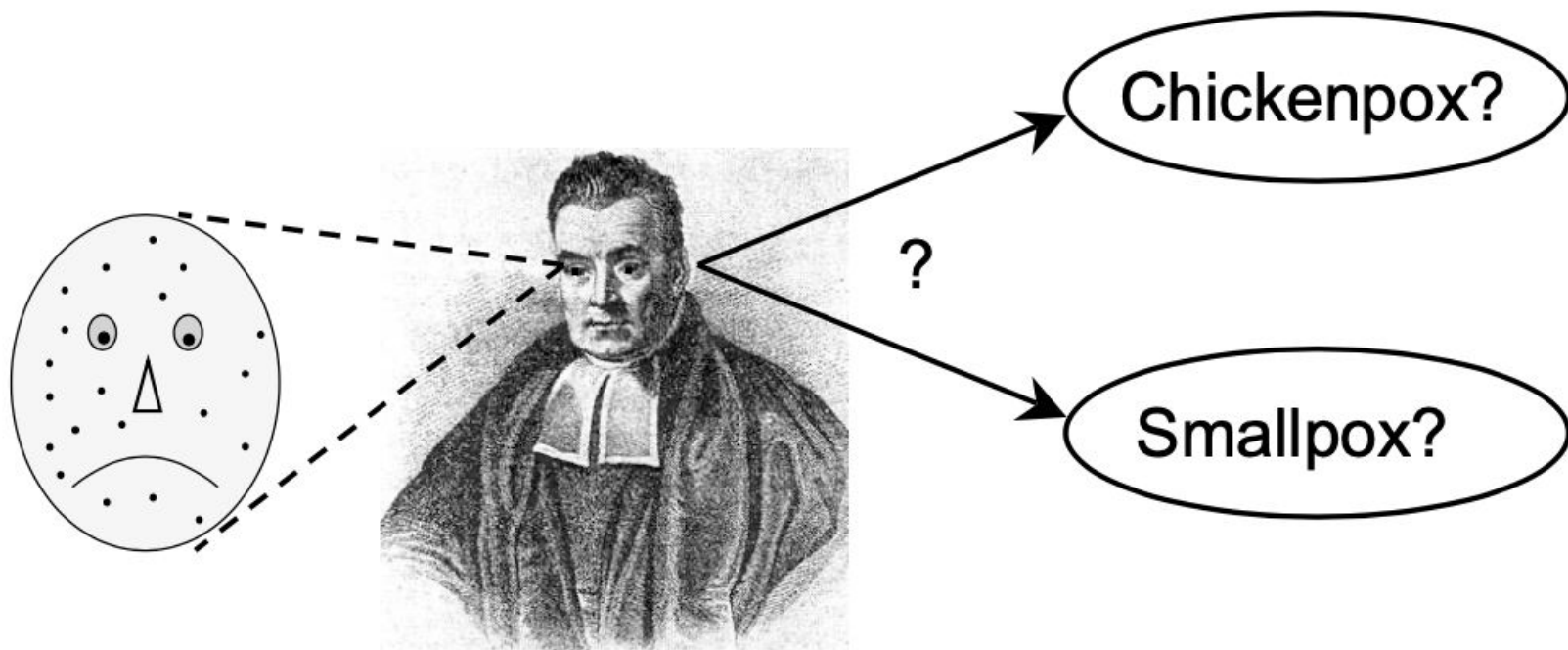
Exploring Distributions and Associations



Motifs aid transparent reasoning and XAI



Bayes Rule



We know,

$$p(\text{spots}|\text{smallpox}) = 0.9.$$

$$p(\text{spots}|\text{chickenpox}) = 0.8.$$

**When You Hear Hooves, Think Horse,
Not Zebra**



Likelihood

$$p(\text{spots}|\text{smallpox}) = 0.9.$$

Likelihood of smallpox = probability of spots given smallpox

$$p(\text{spots}|\text{chickenpox}) = 0.8.$$

Likelihood of chickenpox = probability of spots given smallpox

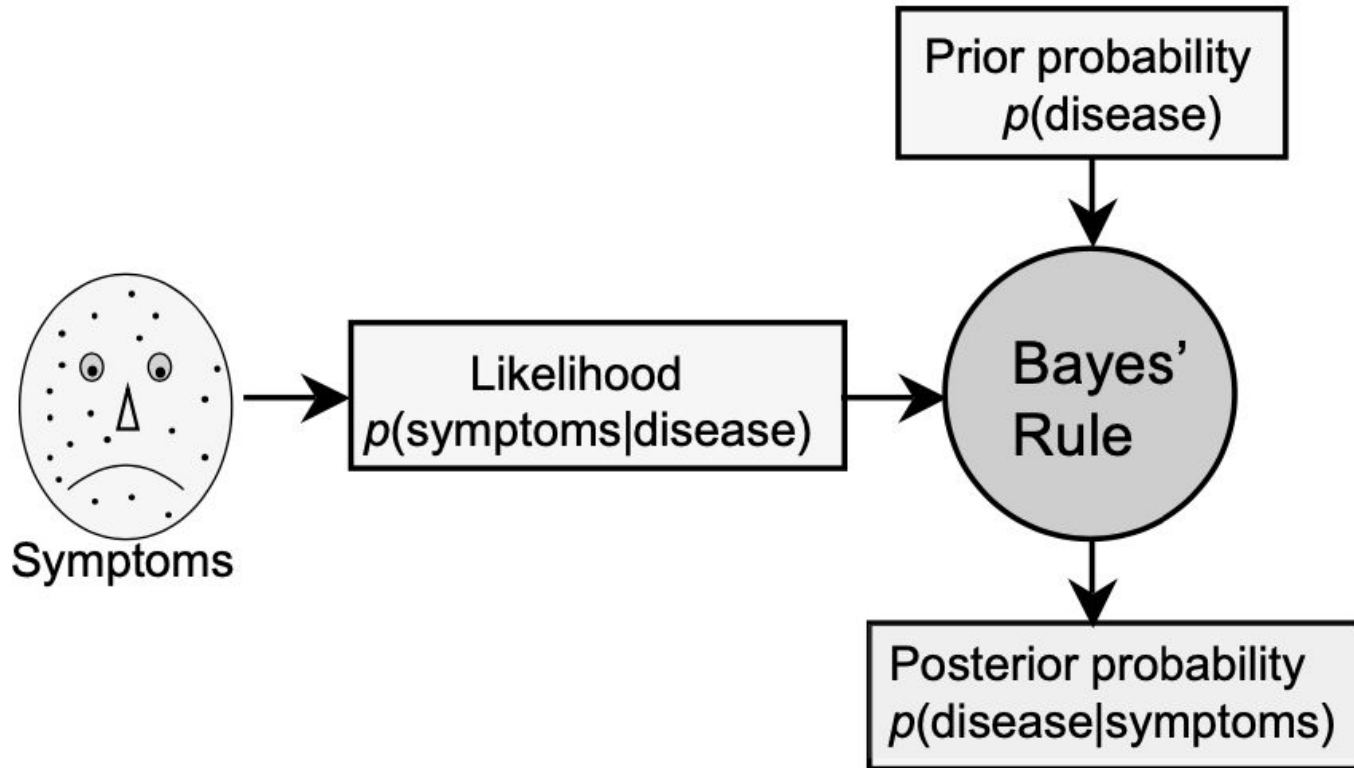
Beware the confusion in language!

Maximum Likelihood Estimate

$$p(\text{spots}|\text{smallpox}) = 0.9. \quad > \quad p(\text{spots}|\text{chickenpox}) = 0.8.$$

Statistical models that work by maximizing the value of likelihood is known as maximum likelihood estimate (MLE)

Bayes Rule in Machine Learning

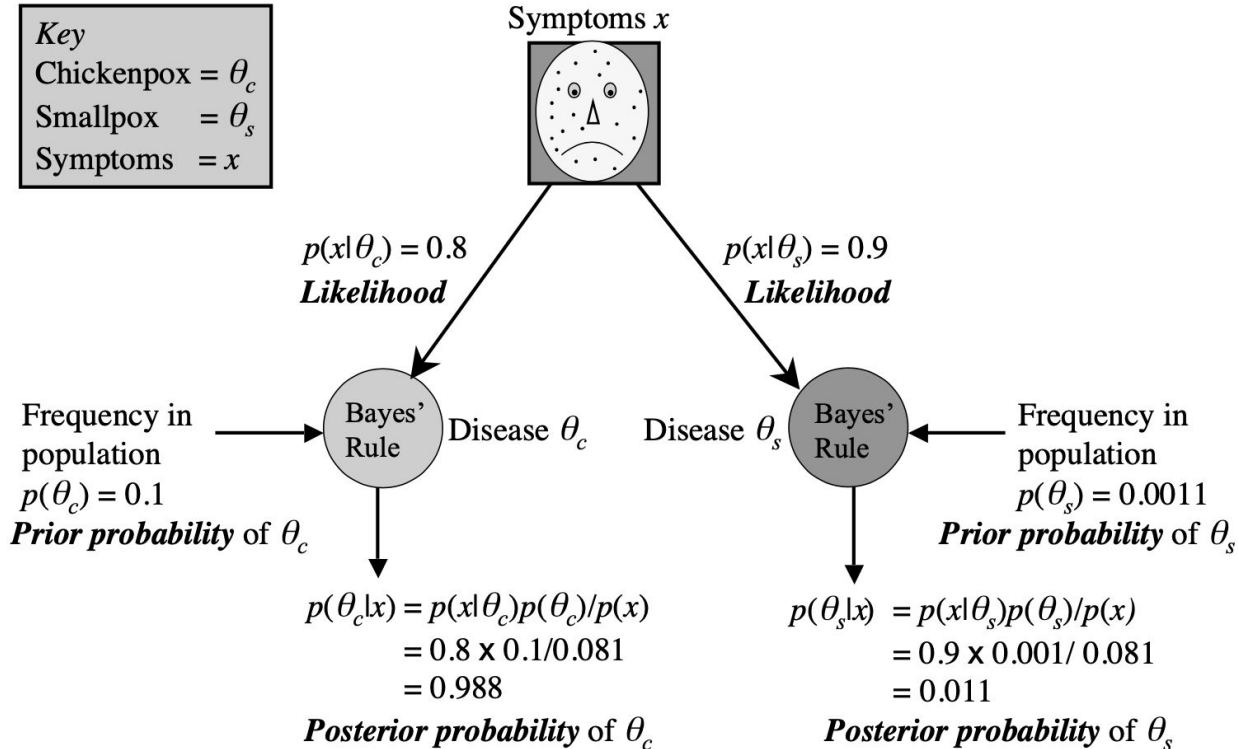


Bayes Rule

$$p(\text{smallpox}|\text{spots}) = \frac{p(\text{spots}|\text{smallpox}) \times p(\text{smallpox})}{p(\text{spots})}.$$

```
# likelihood = prob of spots given smallpox.
pSpotsGSmallpox <- 0.9;
# prior = prob of smallpox.
pSmallpox <- 0.001;
# marginal likelihood = prob of spots.
pSpots <- 0.081;
# find posterior = prob of smallpox given spots.
pSmallpoxGSpots = pSpotsGSmallpox * pSmallpox / pSpots;
#print
s <- sprintf("pSmallpoxGSpots = %.3f", pSmallpoxGSpots)
print(s)
# Output:    pSmallpoxGSpots = 0.011
```

Bayesian Inference



Maximum a Posteriori (MAP) Estimate

Smallpox

$$p(\theta_s|x) = \frac{p(x|\theta_s) \times p(\theta_s)}{p(x)}.$$

Chickenpox

$$p(\theta_c|x) = \frac{p(x|\theta_c) \times p(\theta_c)}{p(x)}.$$

Succinct Notation

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}.$$

$$p(\text{hypothesis}|\text{data}) = \frac{p(\text{data}|\text{hypothesis}) \times p(\text{hypothesis})}{p(\text{data})}$$

Maximum a Posteriori (MAP) Estimate

Smallpox

$$p(\theta_s|x) = \frac{p(x|\theta_s) \times p(\theta_s)}{p(x)}.$$

Chickenpox

$$p(\theta_c|x) = \frac{p(x|\theta_c) \times p(\theta_c)}{p(x)}.$$

$$R_{post} = \frac{p(x|\theta_c)}{p(x|\theta_s)} \times \frac{p(\theta_c)}{p(\theta_s)}.$$

What has cancelled out?

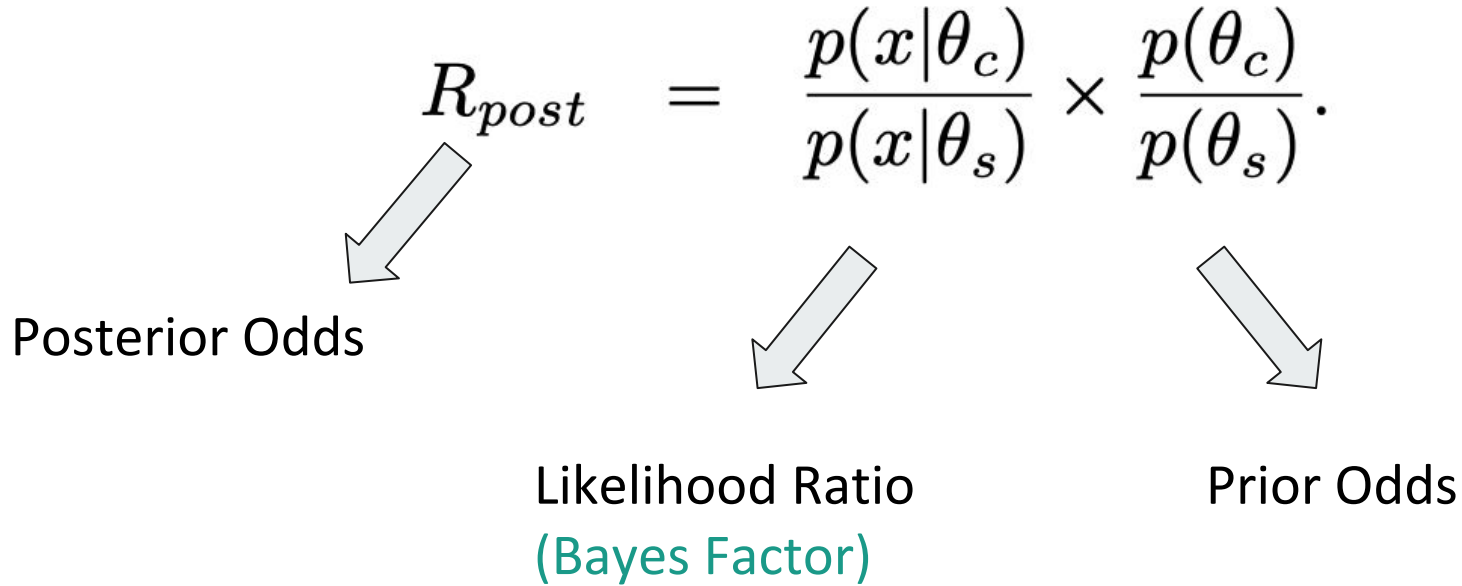
Bayes Factor

$$R_{post} = \frac{p(x|\theta_c)}{p(x|\theta_s)} \times \frac{p(\theta_c)}{p(\theta_s)}.$$

Posterior Odds

Likelihood Ratio
(Bayes Factor)

Prior Odds



The cancelled out term is also called Marginal Likelihood or Evidence

Model Selection

posterior odds = Bayes factor $\times$ prior odds.

$$R_{post} = \frac{0.80}{0.90} \times \frac{0.1}{0.001} = 88.9.$$

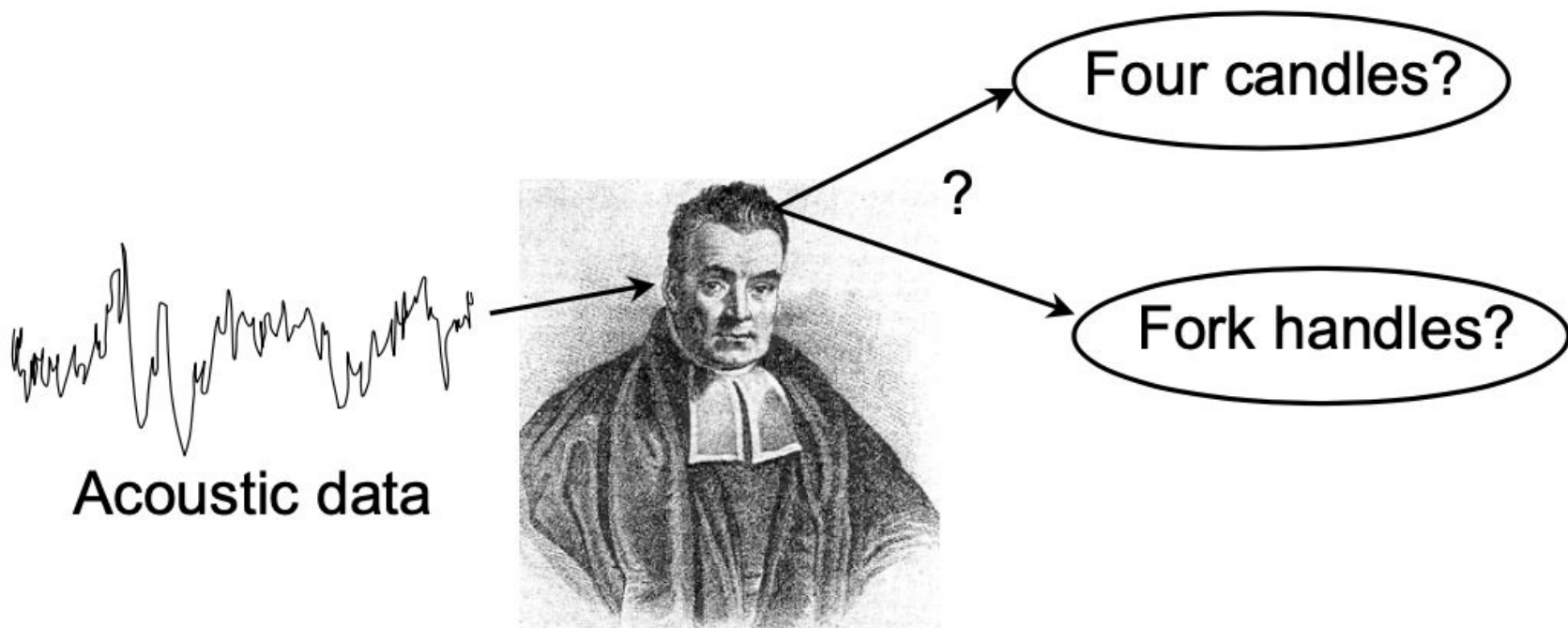
Thumb rule: A posterior odds greater than 3 or less than 1/3 is considered substantial difference between the probabilities of the two models.

When is Bayes Rule Useful?



<https://www.youtube.com/watch?v=pV1IP4N9ajg>

ML with Bayesian Models



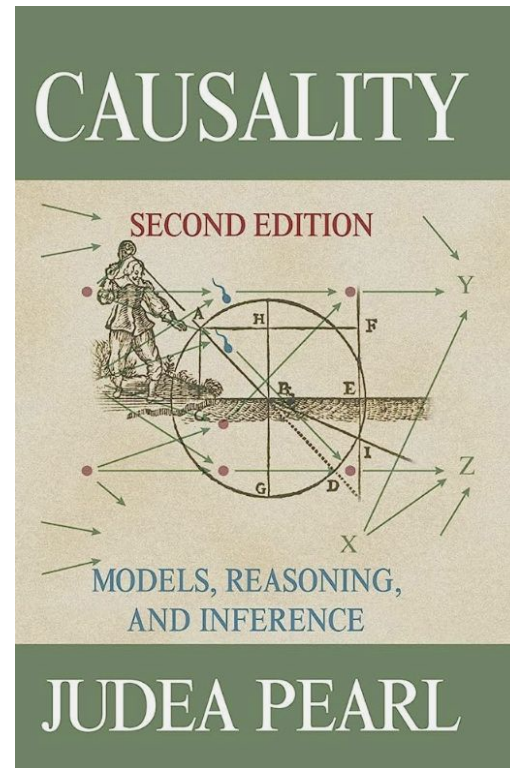
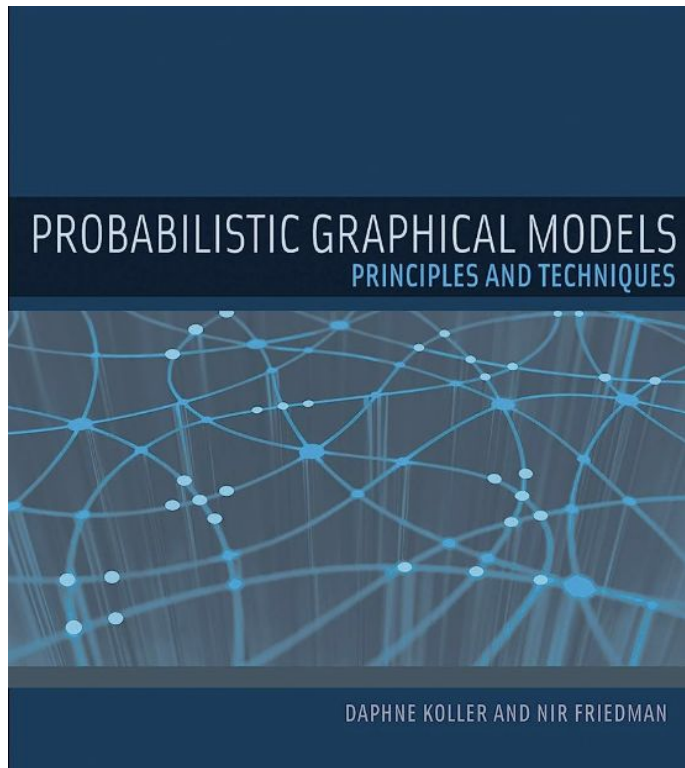
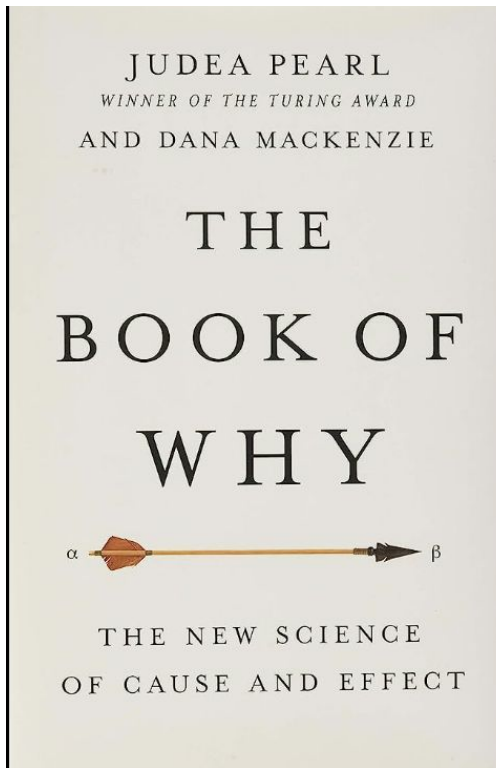
TOWARDS BAYESIAN NETWORKS

Learning Objectives

Participants should be better able to *understand* and *implement*

- Probabilistic reasoning with complex datasets.
- Causal reasoning in complex datasets with interventions
- Explainable Artificial Intelligence models with Bayesian networks learned with *wiseR*
- Decision making under uncertainty
- Deploy models as web-applications with *wiseR*

Recommended Reading

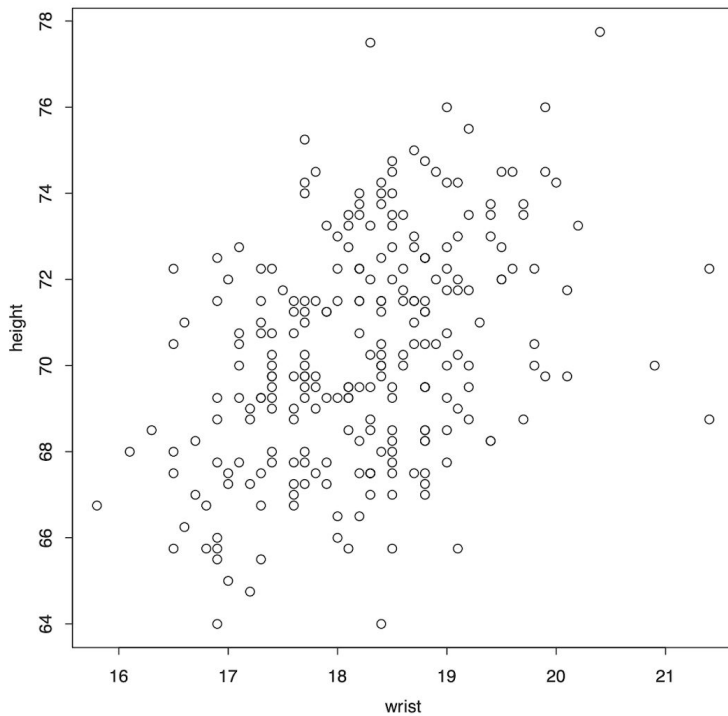
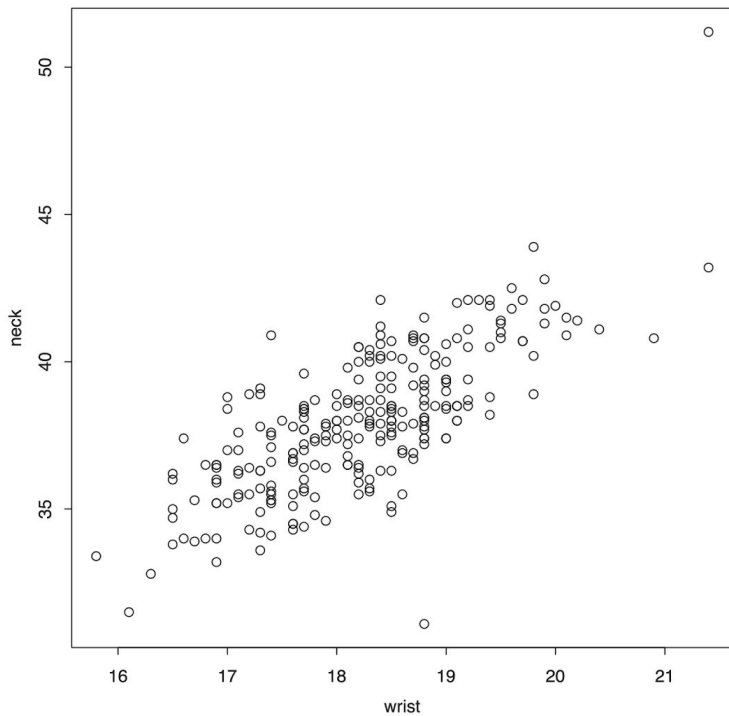


Association: Thinking with Correlations

$$\text{cov}(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y}).$$

$$\text{cor}(x, y) = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \text{cov}(x, y) / (s_x s_y).$$

Correlations is not Causation



Anscombe's Quartet

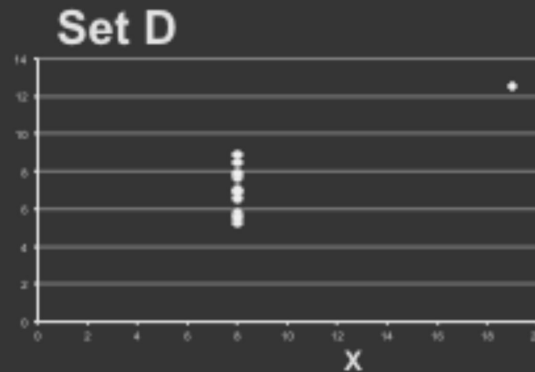
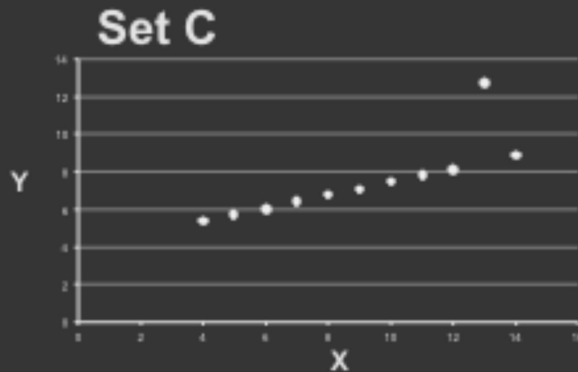
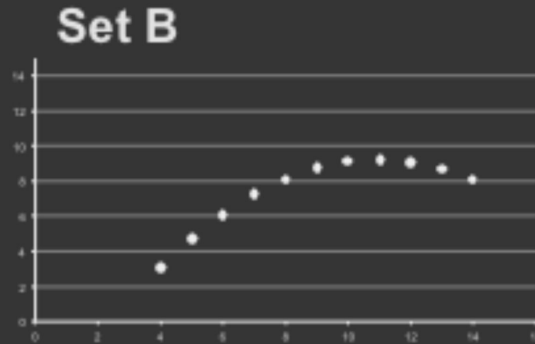
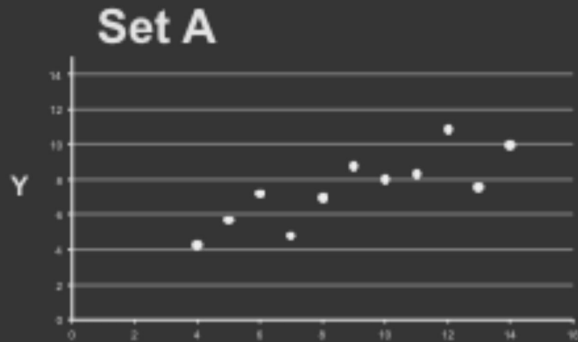
Set A		Set B		Set C		Set D	
X	Y	X	Y	X	Y	X	Y
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.11	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

Summary Statistics Linear Regression

$u_X = 9.0$ $\sigma_X = 3.317$ $Y = 3 + 0.5 X$
 $u_Y = 7.5$ $\sigma_Y = 2.03$ $R^2 = 0.67$

[Anscombe 73]

Anscombe's Quartet



Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing

Justin Matejka and George Fitzmaurice
Autodesk Research, Toronto Ontario Canada
{first.last}@autodesk.com

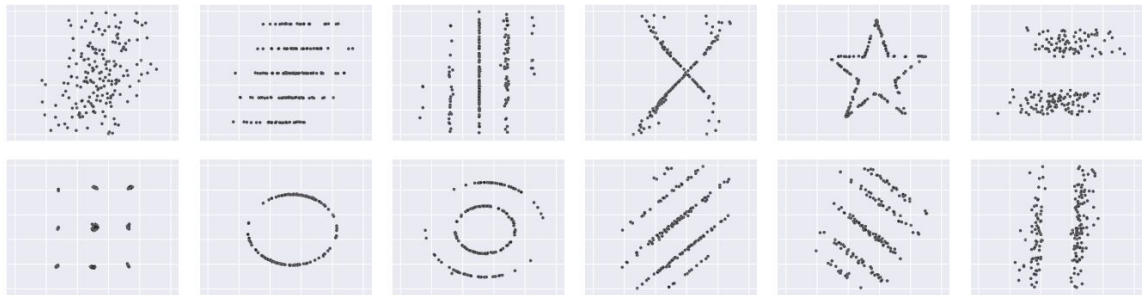
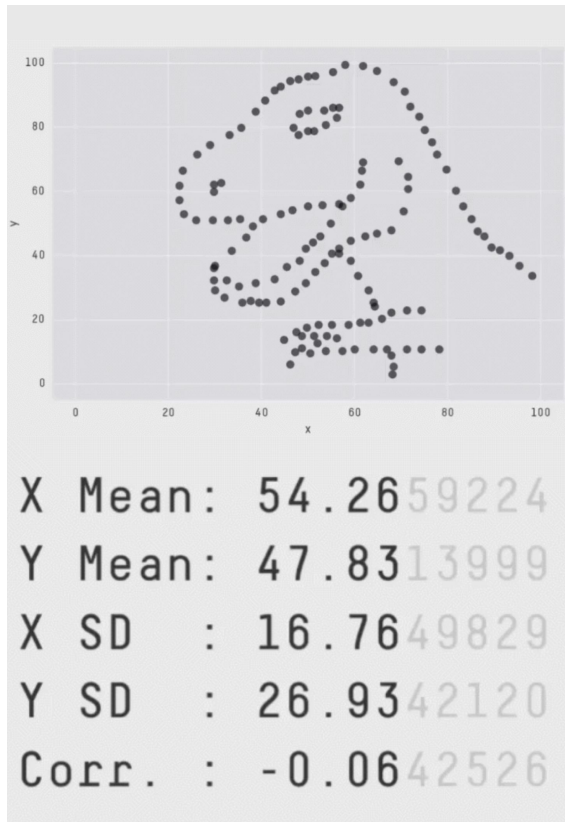
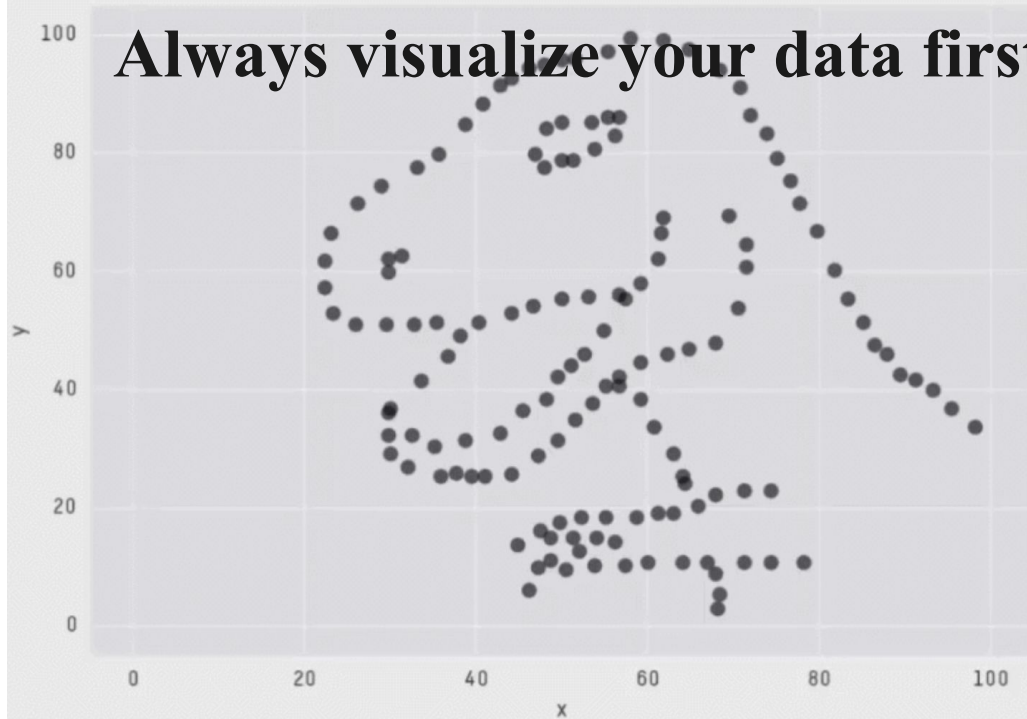


Figure 1. A collection of data sets produced by our technique. While different in appearance, each has the same summary statistics (mean, std. deviation, and Pearson's corr.) to 2 decimal places. ($\bar{x}=54.02$, $\bar{y}=48.09$, $sd_x=14.52$, $sd_y=24.79$, Pearson's $r=+0.32$)



Always visualize your data first!



X Mean: 54.2659224

Y Mean: 47.8313999

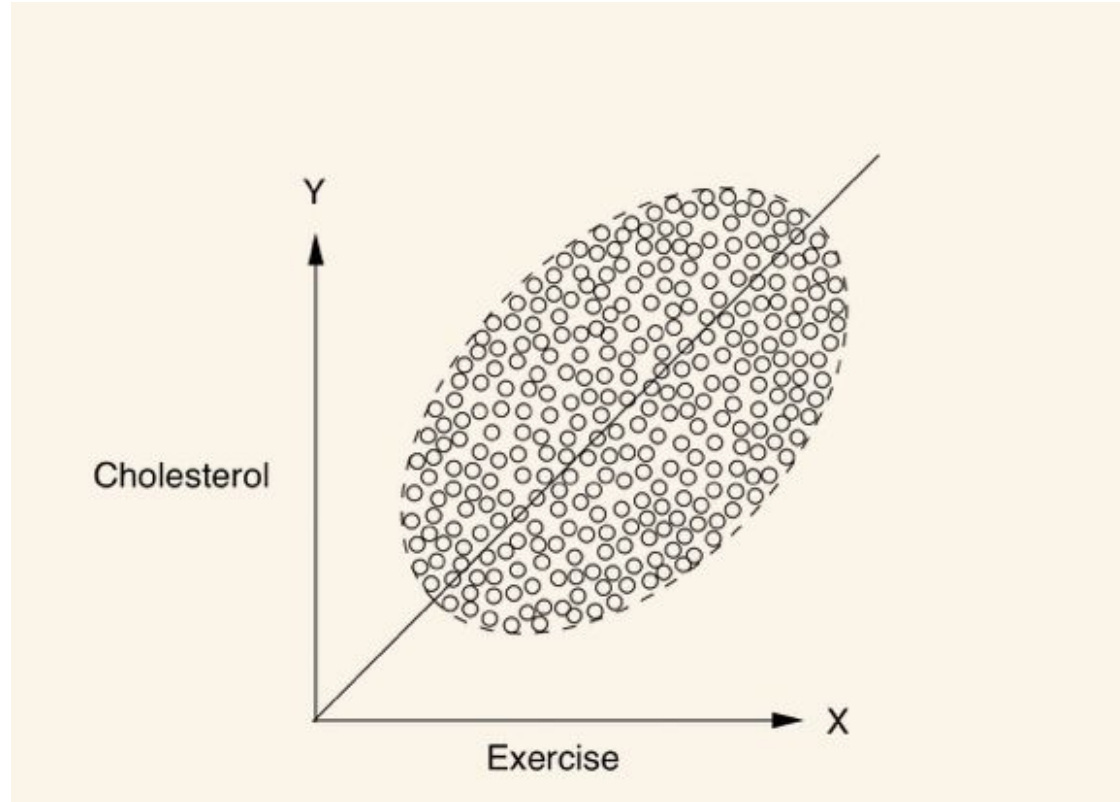
X SD : 16.7649829

Y SD : 26.9342120

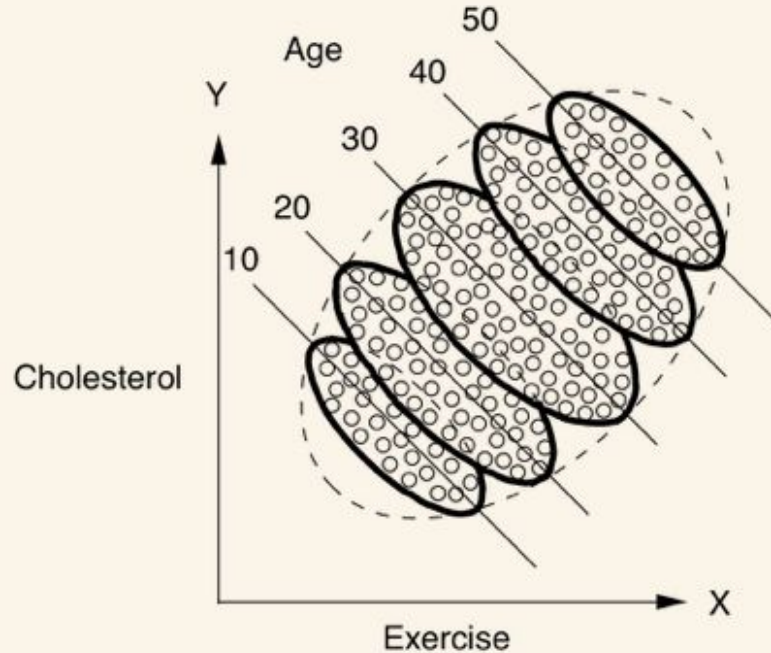
Corr. : -0.0642526

Confounding

Correlations and Confounding



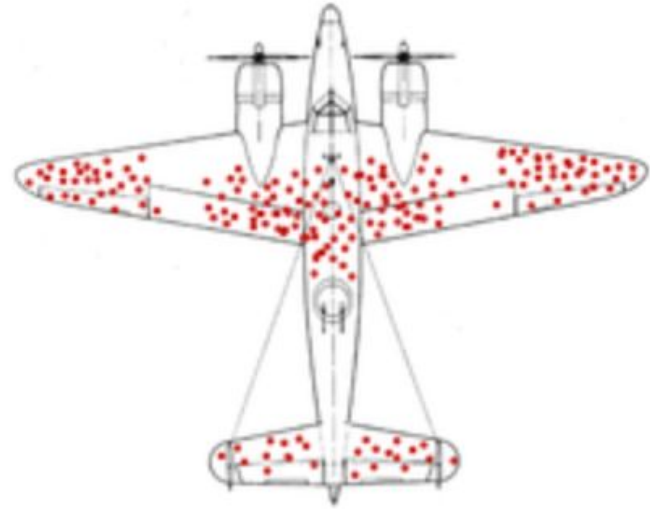
Hidden Confounding Factors



<http://bayes.cs.ucla.edu/PRIMER/>

Correlation is not causation!

Where to put the armor?



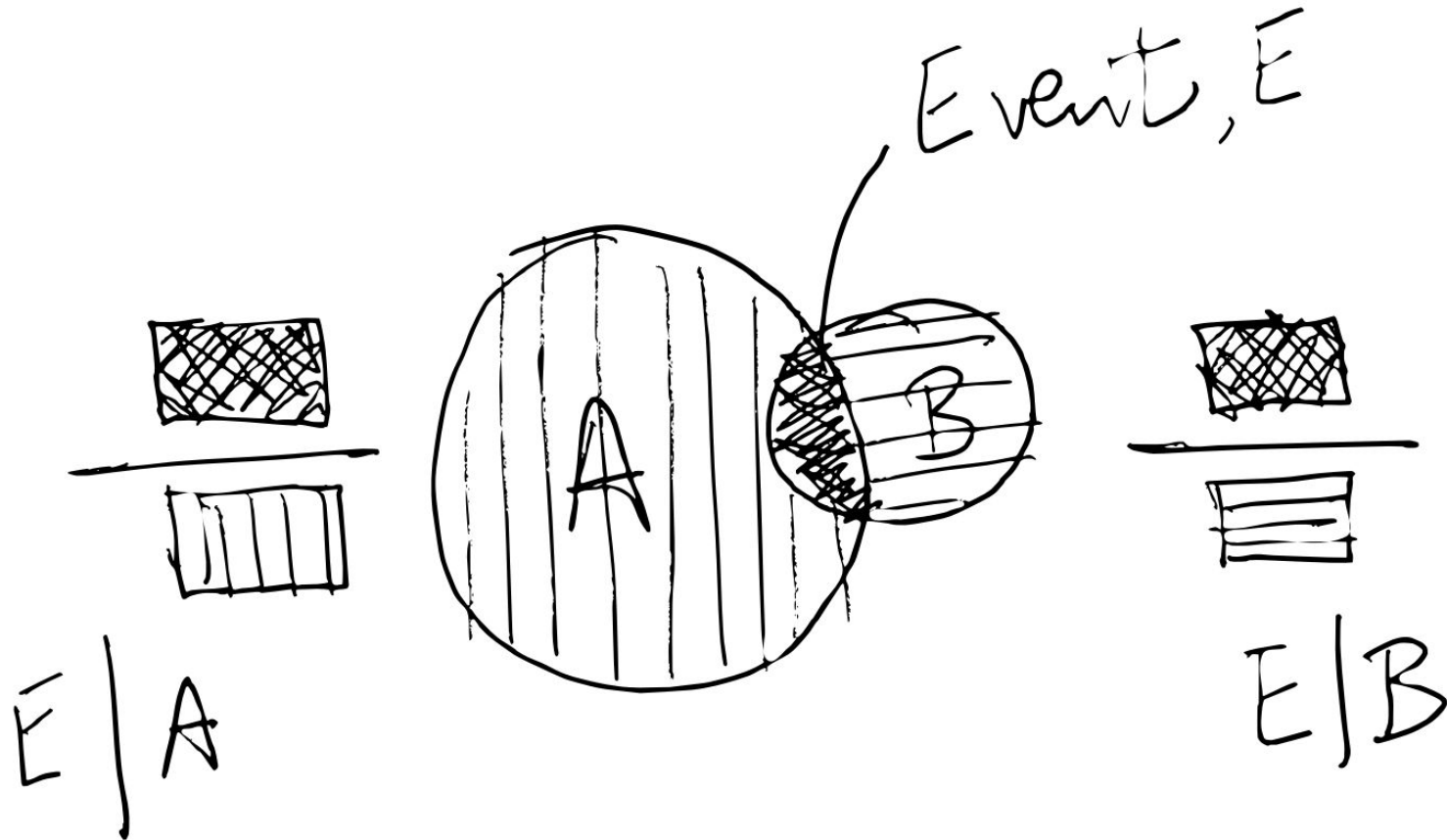
Israeli Data: August 15, 2021

From: <https://datadashboard.health.gov.il/COVID-19/general>

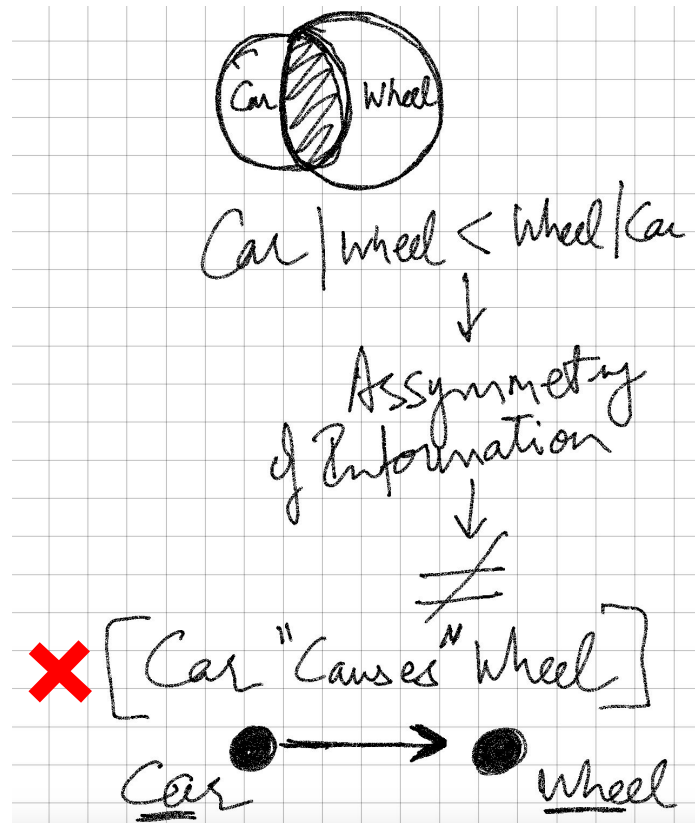
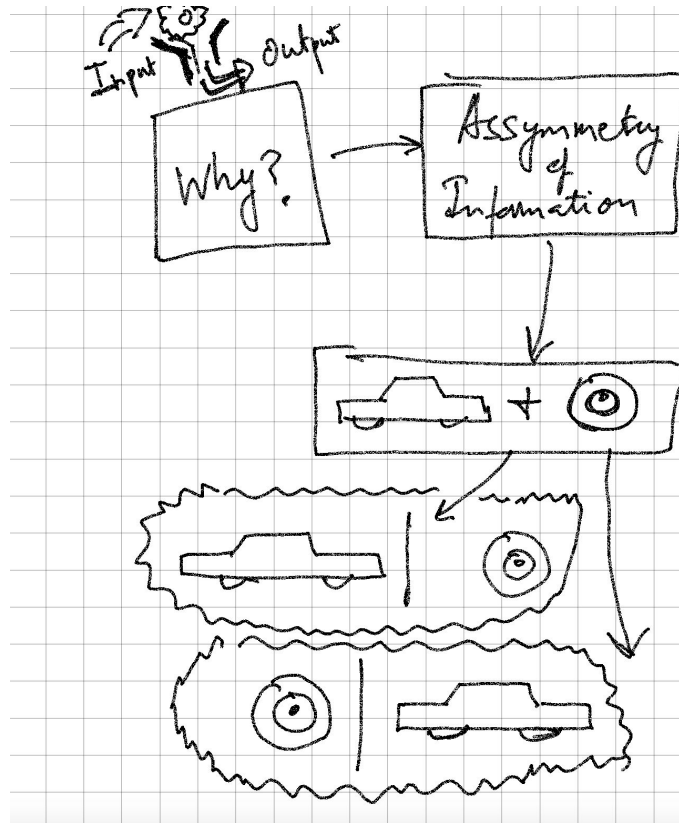
Age	Population (%)		Severe cases		Efficacy
	Not Vax %	Fully Vax %	Not Vax per 100k	Fully Vax per 100k	vs. severe disease
All ages	1,302,912 18.2%	5,634,634 78.7%	214 16.4	301 5.3	67.5%
<50	1,116,834 23.3%	3,501,118 73.0%	43 3.9	11 0.3	91.8%
>50	186,078 7.9%	2,170,563 90.4%	171 90.9	290 13.6	85.2%

This strange result illustrates something called **Simpson's Paradox**, in this case meaning you can have **very high efficacy in each group**, but the **overall efficacy looks much lower** because one group (older people) is **more vaccinated** and have a **much higher risk of severe disease**.

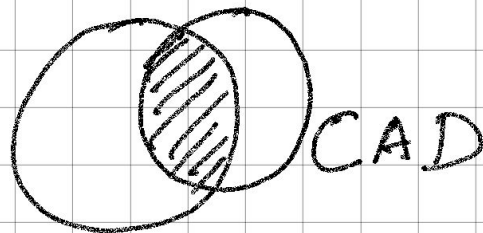
Conditional Probability



Bayes Rule Exploits Real World Asymmetry

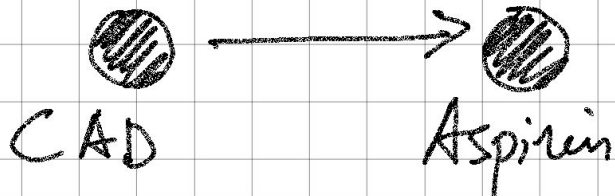


Another Example

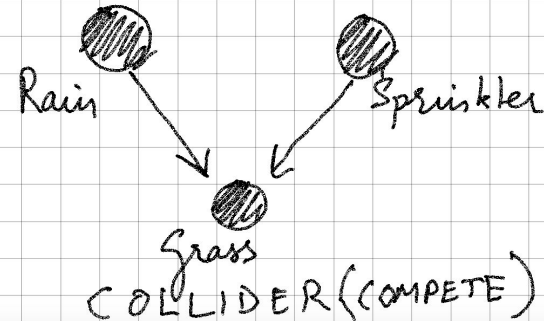
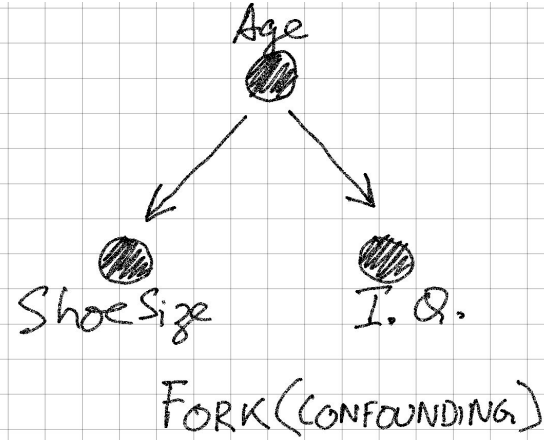
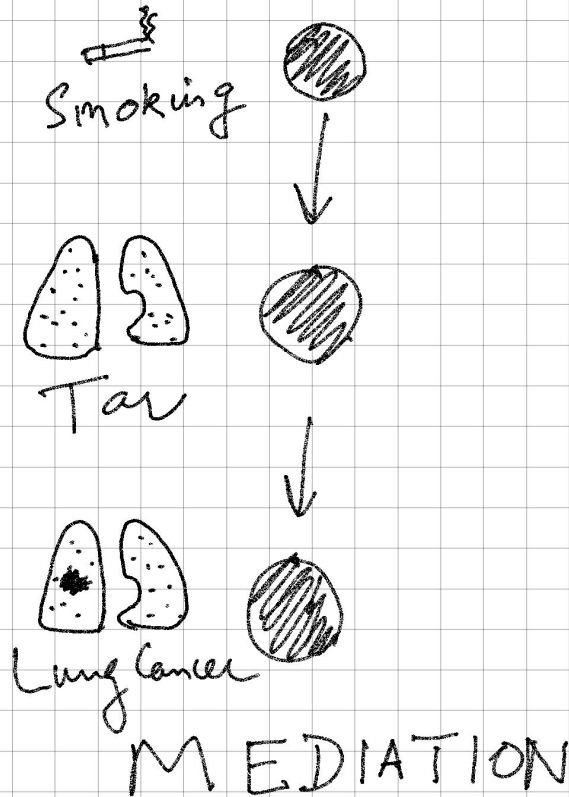


Aspirin

$$P_r(\text{Aspirin} / \text{CAD}) > P_r(\text{CAD} / \text{Aspirin})$$



World of Conditional Probabilities



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- Causal reasoning in complex datasets with interventions
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- Decision making under uncertainty
- Deploy models as web-applications with *wiseR*

Key Steps in Building a BN Model

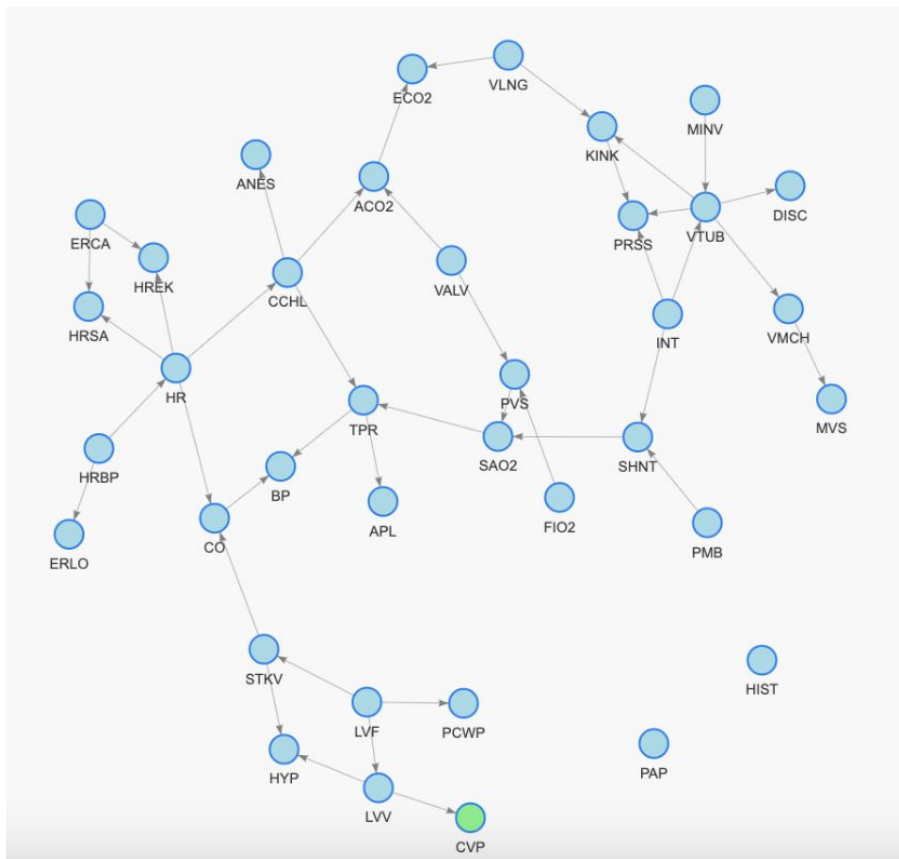
- Learn Structure
 - Score Based
 - Constraint Based
- Validate Structure
 - Bootstrapping
 - Domain Based Sanitization
- Conduct Inference
 - Exact Inference
 - Approximate Inference

Example Set up

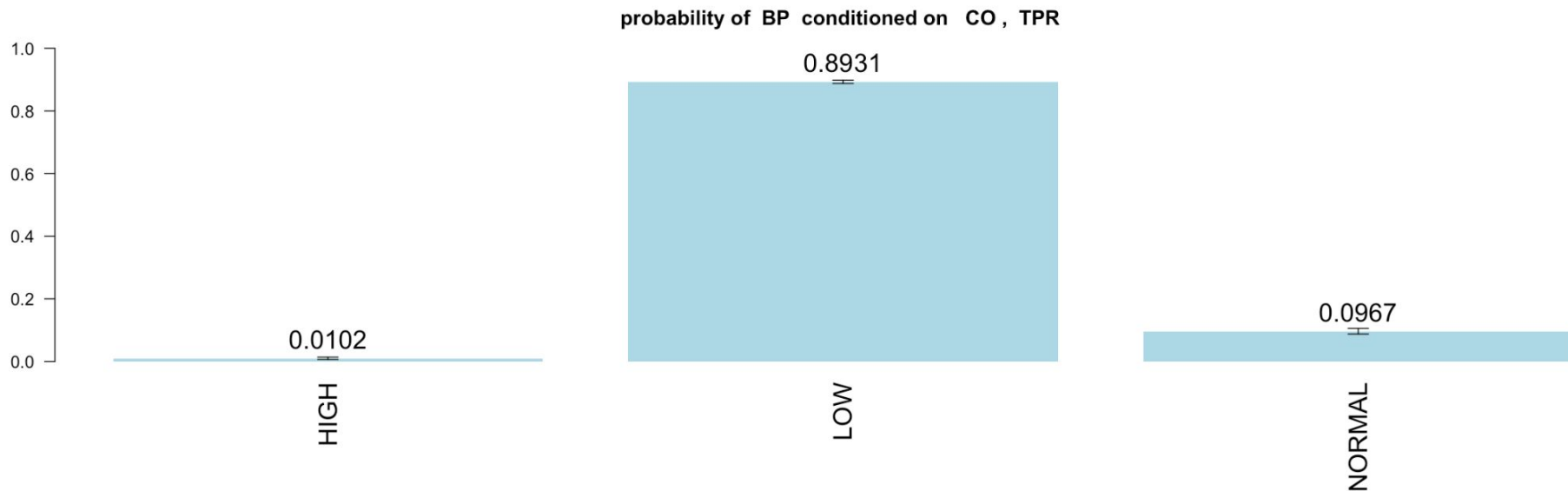
We learn the Network structure using the following parameters:

- Algorithm: Hill Climbing
- Network Score: Akaike Information Criterion
- Parameter Fitting: Bayesian Parameter Estimation
- Blacklisting: No
- Bootstrap Model: Yes
- No. of Bootstrap: 101
- Edge Strength: >0.51
- Direction Strength: >0.51
- Sample Portion: 1

Learning Structure (ICU Alarms Example)



Conducting Inference



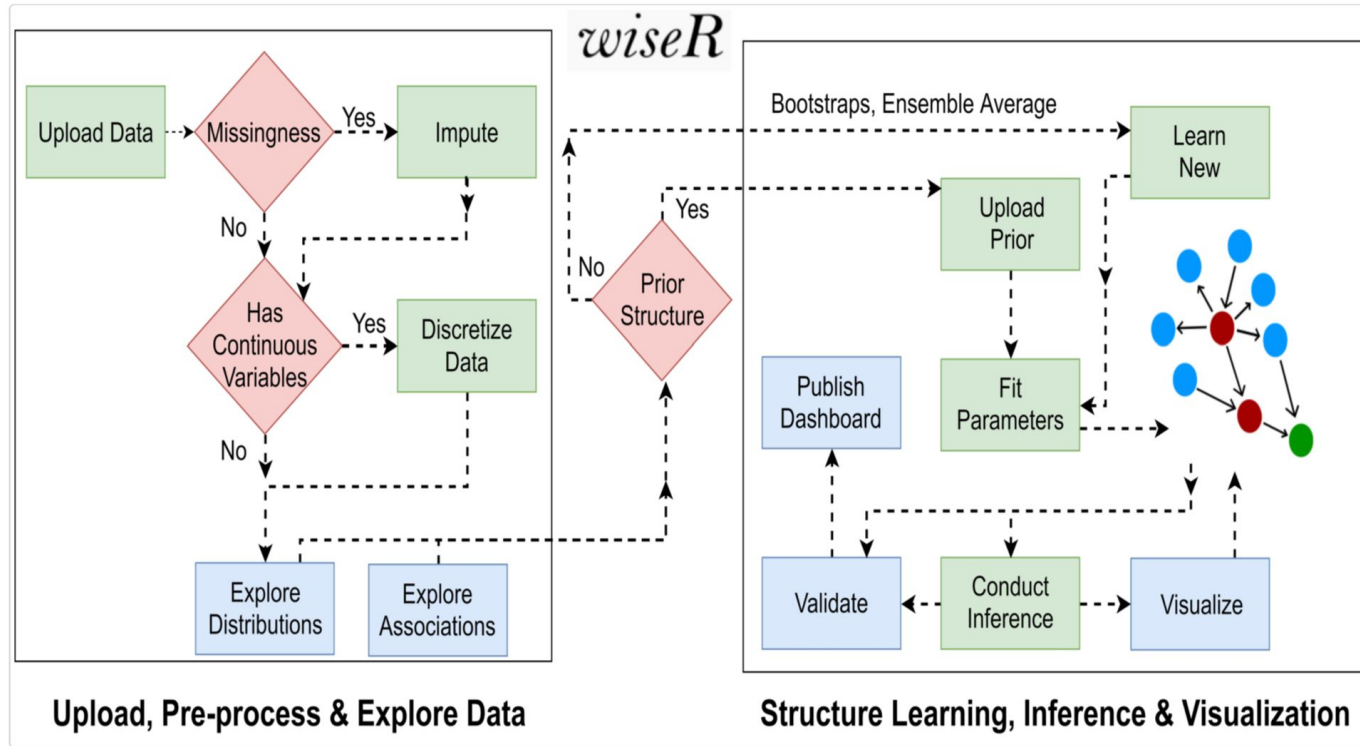
Probabilistic Reasoning

Examples: Anaphylaxis and Total Peripheral Resistance (APL, TPR). Baseline rate of APL: 1%, knowing that the patient had low TPR changes our “probabilistic belief” to 3%

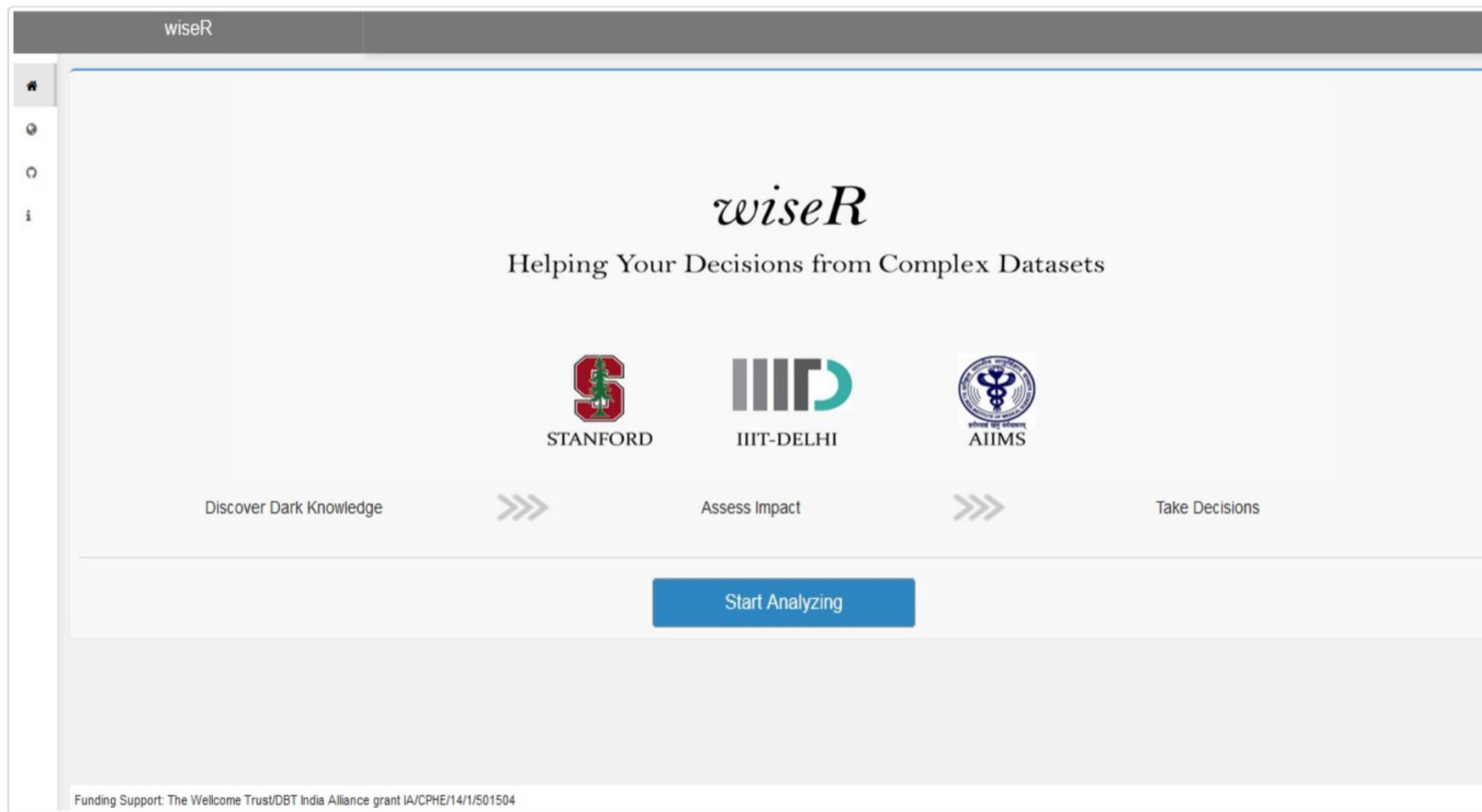
Motifs:

1. Mediation effect: CCHL $\rightarrow$ MINV ... $\rightarrow$ ACO2
2. Confounding effect
3. Inter-causal reasoning: CO $\rightarrow$ BP $\leftarrow$ TPR
 - CO $\leftarrow$ BP $\rightarrow$ HR

wiseR: Bayesian Artificial Intelligence



wiseR: Bayesian Artificial Intelligence



The image shows a screenshot of the wiseR web application interface. At the top, there is a dark grey header bar with the text "wiseR" in white. Below the header, on the left side, is a vertical sidebar containing a home icon, a magnifying glass icon, a circular arrow icon, and a lowercase 'i' icon. The main content area has a light grey background. In the center, the word "wiseR" is written in a large, elegant, italicized serif font. Below it, the tagline "Helping Your Decisions from Complex Datasets" is displayed in a smaller, sans-serif font. Further down, three logos are arranged horizontally: the Stanford University logo (a red block 'S' with a green redwood tree in front of it), the IIT-Delhi logo (the letters "IIITD" in a stylized grey and teal font), and the All India Institute of Medical Sciences (AIIMS) logo (a circular emblem with a caduceus in the center and text in Hindi and English). Below these logos, a horizontal flow of three steps is shown: "Discover Dark Knowledge" followed by a double right-pointing arrow, "Assess Impact" followed by another double right-pointing arrow, and "Take Decisions". At the bottom center of the main content area is a prominent blue rectangular button with the white text "Start Analyzing". At the very bottom of the page, a small line of text reads: "Funding Support: The Wellcome Trust/DBT India Alliance grant IA/CPHE/14/1/501504".

wiseR

wiseR

Helping Your Decisions from Complex Datasets


STANFORD


IIIT-DELHI


AIIMS

Discover Dark Knowledge >>> Assess Impact >>> Take Decisions

Start Analyzing

Funding Support: The Wellcome Trust/DBT India Alliance grant IA/CPHE/14/1/501504

Practical Session

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wiseR

App Settings Data Association Network Bayesian Network Decision Network Publish your dashboard App Tutorial

Enable Parallel Computing

Number of clusters

1

Loading Data

wiseR

App Settings

Data

Association Network

Bayesian Network

Decision Network

Publish your dashboard

App Tutorial

Dataset

Explore

upload

Pre-Process

Download

Choose default dataset

Alarm

load

Data Format:

.CSV

variables as Factor:

☒

File Input:

Browse...

No file selected

Search:

HRBP	HREK	HRSA	PAP	SAO2	FIO2	PRSS	ECO2	MINV	MVS	F
HIGH	HIGH	HIGH	NORMAL	NORMAL	LOW	HIGH	ZERO	HIGH	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	HIGH	ZERO	ZERO	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	NORMAL	ZERO	ZERO	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	NORMAL	NORMAL	HIGH	ZERO	ZERO	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	LOW	ZERO	ZERO	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	LOW	NORMAL	HIGH	HIGH	ZERO	NORMAL	FA
HIGH	HIGH	HIGH	NORMAL	LOW	LOW	LOW	ZERO	ZERO	NORMAL	FA
8	NORMAL	NORMAL	FALSE	NORMAL	LOW	LOW	HIGH	NORMAL	NORMAL	FA
9	NORMAL	LOW	FALSE	LOW	LOW	HIGH	HIGH	HIGH	NORMAL	FA
10	NORMAL	NORMAL	FALSE	NORMAL	NORMAL	HIGH	HIGH	HIGH	LOW	FA

Showing 1 to 10 of 20,000 entries

Previous

1

2

3

4

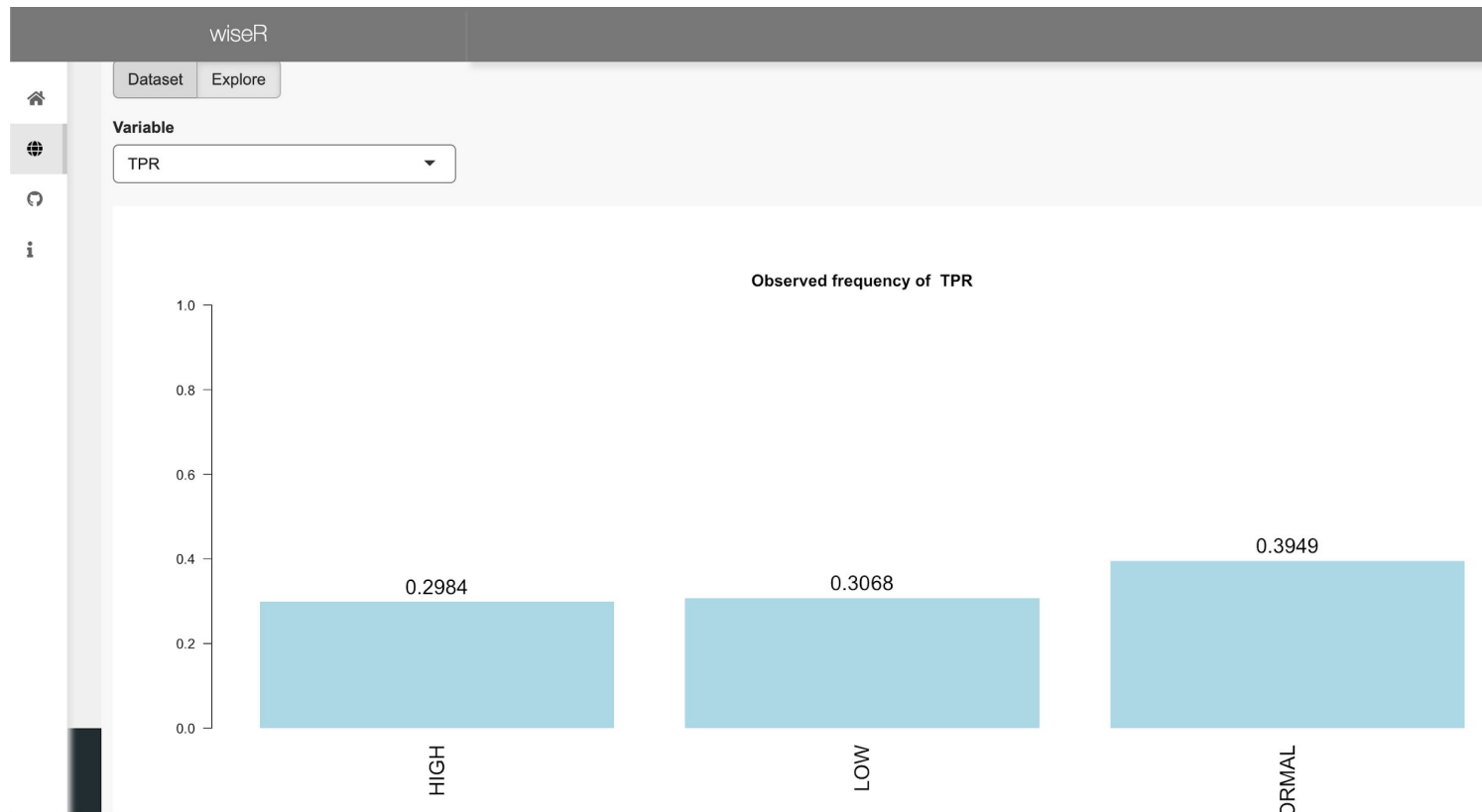
5

...

2000

Next

Exploration of univariate distributions



wiseR

App Settings Data **Association Network** Bayesian Network Decision Network Publish your dashboard App Tutorial

Association Network Export Table

Build Visual Settings Refresh

Nth neighbors(of selection): graph Detect Modules

Select by id

The graph displays a network of entities represented by blue circular nodes. The nodes are interconnected by thin grey lines, representing associations. The entities are labeled with three-letter codes: APL, HRBP, TPRI, BP, AR, CCHL, VMGH, DISC, MVF, VTUB, CO2, ACO2, SHNT, PMB, KINK, INT, PRSS, ALNG, MINT, PVLV, VALV, SAQ2, HRSA, HRQ2, CO, CVP, HST, ATKV, LVP, PCVP, LVA, HYP, and LVF. The network is organized into several clusters, with CCHL acting as a central hub connecting to multiple other nodes. The layout is spread across the width of the interface, with some nodes positioned higher or lower to avoid overlap.

Motifs aid transparent reasoning and XAI

